Ain Shams University

Faculty of Computer & Information Sciences

Computer Science Department

MyHR — An Adaptive AI-Powered Hiring & Interview Platform

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June 2026

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MyHR — An Adaptive AI-Powered Hiring & Interview Platform

This documentation is submitted in partial fulfillment of the requirements for the **Bachelor's degree in Computer and Information Sciences**

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0.1	2026-06	Project Team	Initial draft of all chapters
1.0	2026-06	Project Team	First complete release for review

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Finally, we thank our friends and everyone who supported us throughout the development of this project.

Abstract

Recruitment at scale is bottlenecked by two manual, time-consuming, and inconsistency-prone tasks: screening large volumes of résumés (CVs) against a job description (JD), and conducting first-round interviews. **MyHR** is an adaptive, AI-powered hiring platform that automates both. It pairs an **enterprise hiring portal** — where companies post jobs, upload CVs in batches, and obtain neural rankings of candidates — with an **AI interview engine** that conducts a grounded, adaptive, voice-or-text interview and produces a defensible, server-computed score for every candidate.

The platform is built on three cooperating subsystems. A **Large Language Model (LLM) agent**, orchestrated as a state machine with LangGraph and powered by Groq's `llama-3.3-70b-versatile`, generates interview questions and evaluates answers. A **hybrid Retrieval-Augmented Generation (RAG) layer** grounds every question in the candidate's own CV and the JD using sparse retrieval (BM25), dense retrieval (Pinecone with `all-mpnet-base-v2` embeddings), Reciprocal Rank Fusion, and a cross-encoder reranker. A **custom neural layer** of eight PyTorch models contributes CV–JD skill matching, multi-dimensional answer evaluation, candidate ranking, adaptive question difficulty (reinforcement learning), emotion analysis, and performance prediction. Final answer scores are produced by a transparent blend that weights the LLM judgment at 65%, the neural evaluator at 20%, and the deep scoring model at 15%, with a relevance gate that suppresses the neural contribution on off-topic answers.

The system is delivered as a **React** single-page application and a **FastAPI** backend, with **Firebase Authentication**, **Cloud Firestore** persistence, role-based access control separating candidates from enterprise HR users, and token-based public interview links. This document describes the problem, the relevant background, the proposed system architecture, the detailed implementation of each component, the testing strategy, and the project's conclusions and future work.

Keywords: AI interview, retrieval-augmented generation, LLM agent, candidate ranking, neural scoring, FastAPI, React, Firestore, adaptive difficulty.

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List of Abbreviations

Abbreviation	Meaning
API	Application Programming Interface
BM25	Best Matching 25 (sparse ranking function)
CV	Curriculum Vitae (résumé)
HR	Human Resources
JD	Job Description
LLM	Large Language Model
MLP	Multi-Layer Perceptron
NDCG	Normalized Discounted Cumulative Gain
PII	Personally Identifiable Information
PPO	Proximal Policy Optimization
RAG	Retrieval-Augmented Generation
RBAC	Role-Based Access Control
RL	Reinforcement Learning
RRF	Reciprocal Rank Fusion
SPA	Single-Page Application
STT	Speech-to-Text
TTS	Text-to-Speech
WS	WebSocket

Chapter One

Introduction

Chapter Outline

1.1 Problem Definition

Hiring is one of the most resource-intensive processes a company undertakes, and two of its earliest stages are also its most repetitive:

- **Résumé screening.** For a single open role, recruiters may receive hundreds of CVs. Each must be read, compared against the job description, and judged for fit. The process is slow, subjective, and inconsistent between reviewers and across time of day.
- **First-round interviewing.** Even after shortlisting, conducting an initial screening interview for every promising candidate consumes scarce interviewer time. Scheduling alone introduces days of delay, and the quality of questions varies with the interviewer's preparation and fatigue.

These two stages share a common weakness: they depend on a human reading the same kinds of documents and asking the same kinds of questions, over and over, with no guarantee of consistency. The result is a hiring funnel that is **slow, expensive, hard to audit, and vulnerable to unconscious bias**.

MyHR addresses this problem directly. It provides:

1. An **enterprise portal** that ingests CVs in bulk, parses them, matches them against the job's required skills, and ranks candidates with neural models — turning hundreds of documents into an ordered shortlist in seconds.

2. An **AI interview engine** that conducts an adaptive, grounded interview with each invited candidate and returns a single, defensible score and a structured report, without occupying any human interviewer's time.

In short, the problem statement is: *manual CV screening and first-round interviewing are slow, inconsistent, and costly; they can be made faster and more consistent by grounding an LLM-driven interviewer in the candidate's own documents and by scoring candidates with purpose-trained neural models.*

1.1.1 History

The two enabling technologies behind MyHR matured only recently:

- **Applicant Tracking Systems (ATS)** have existed for decades, but classical ATS rely on keyword matching. They reward candidates who echo the job description's wording rather than candidates who actually possess the underlying skills, and they cannot conduct an interview.
- **Large Language Models (LLMs)** became capable of fluent, context-aware question generation and answer evaluation only after the transformer architecture and instruction-tuned chat models. However, an LLM used naively will *hallucinate* — it may ask about experience the candidate never claimed, or score an answer it never actually grounded in the role.
- **Retrieval-Augmented Generation (RAG)** emerged as the standard technique to keep an LLM grounded in trustworthy source documents. By retrieving the most relevant passages of a CV and JD and feeding them to the model, RAG constrains the interview to facts that actually exist in the candidate's profile.

MyHR combines these threads: it replaces keyword ATS matching with neural skill matching, and it replaces an ungrounded LLM with a RAG-grounded interview agent whose scores are cross-checked by dedicated neural evaluators.

1.1.2 Applications

The platform serves two distinct audiences from one codebase:

- **Enterprise hiring (primary).** A company requests access, is approved by a platform administrator, posts jobs, uploads candidate CVs, receives a neural ranking, invites the top candidates to an AI interview by email, and reviews completed interviews and aggregate analytics on a dashboard.
- **Candidate self-practice (secondary).** An individual candidate can sign up and run mock AI interviews against their own CV to prepare for real interviews, tracking their readiness over time.

Both audiences are gated by **role-based access control**: an account is either a *candidate* or an enterprise *HR* user, and the two roles are mutually exclusive for a given email.

1.2 Motivation

The motivation for MyHR is to demonstrate that a **single, coherent system can automate the top of the hiring funnel without sacrificing trustworthiness**. Three concerns drove the design:

- **Grounding over fluency.** An interviewer that sounds fluent but invents facts is worse than useless. MyHR's central engineering bet is that every question and every score must be grounded in retrieved evidence from the candidate's own CV and the JD.
- **Defensibility over convenience.** A candidate's score must be computed on the server from evidence the candidate cannot tamper with, and it must be explainable as a transparent blend of an LLM judgment and purpose-trained neural models rather than a single opaque number.
- **Separation of concerns.** The enterprise workflow (multi-tenant, authenticated, persisted) and the AI interview workflow (stateful, real-time, model-heavy) are genuinely different problems, and the architecture keeps them in separate, well-defined layers.

1.3 Objectives

The project's objectives are grouped into five categories.

Functional objectives — what the system must *do*:

1. Allow companies to onboard (request access → admin approval → invitation) and manage job postings.
2. Ingest candidate CVs in bulk, parse them, and rank candidates against each job.
3. Conduct an adaptive, voice-or-text AI interview with every invited candidate.
4. Produce a per-candidate interview report and a combined hiring score, and present hiring analytics to HR users.

Technical objectives — how the system must be *engineered*:

5. Implement a clean three-layer architecture (enterprise / interview-AI / training) with a FastAPI backend, a React single-page application, and Firestore persistence.
6. Expose a well-defined REST + WebSocket API and cover the core logic with automated tests.
7. Keep the codebase maintainable and containerized for reproducible deployment.

AI objectives — the intelligence the system must exhibit:

8. Ground every interview question in retrieved evidence from the candidate's CV and the JD (hybrid RAG) so the interviewer never hallucinates experience.
9. Score answers through a transparent blend of an LLM judgment and purpose-trained neural models, guarded by a question-to-answer relevance gate.
10. Train and evaluate the neural model layer rigorously, on held-out test sets with standard ranking metrics, experiment tracking, and a human-rating validation study.

Security objectives — the trust guarantees the system must provide:

11. Authenticate all users via Firebase ID tokens and enforce role-based access control that cleanly separates candidates, HR users, and platform administrators.
12. Compute candidate scores server-side only, so a candidate cannot tamper with their result, and issue public interview links as cryptographically strong, expiring tokens.
13. Redact personally identifiable information before any document is placed in the vector store, and sanitize inputs against prompt-injection.

Performance objectives — the responsiveness the system must achieve:

14. Turn a batch of uploaded CVs into a ranked shortlist in seconds rather than hours.
15. Serve hiring analytics from pre-computed aggregates instead of scanning every candidate on each request, and load neural models lazily so application startup is not blocked.

1.4 Scope

In scope. Enterprise account onboarding and approval; job posting; batch CV parsing, skill extraction, and rubric scoring; neural candidate ranking; email-delivered, token-based interview invitations; an adaptive, RAG-grounded, voice/text AI interview with silent proctoring; server-side answer scoring and report synthesis; hiring analytics; candidate self-practice; the full neural training and evaluation pipeline.

Out of scope / partial. Production-grade object storage is stubbed against local disk and a MinIO/S3 configuration rather than a hardened cloud bucket; observability is limited to a health endpoint, an in-process metrics endpoint, and structured logging; the candidate ranker is trained on synthetically generated comparative data pending real labeled outcomes; the human-rating validation study was conducted with a single rater. These items are documented in **Chapter 7 (Known Limitations and Future Work)**.

1.5 Functional Requirements

The functional requirements define the behavior the system must exhibit. **Table 1.2** lists them grouped by actor.

Table 1.2 — Functional Requirements.

ID	Actor	Requirement
FR-1	HR (prospective)	Submit an enterprise access request using a corporate email address.
FR-2	Admin	Review, approve, or reject pending access requests.
FR-3	HR	Accept a company invitation and be granted the HR role.
FR-4	HR	Create, view, edit, close, and reopen job postings.
FR-5	HR	Upload candidate CVs in bulk (PDF/DOCX) for a job.
FR-6	System	Parse each CV, extract skills, match against the JD, and compute a rubric score.
FR-7	HR	View candidates ranked by match score and inspect a per-candidate skill breakdown.
FR-8	HR	Invite a candidate to an AI interview (delivers an email link); blocked when the job is closed.
FR-9	Candidate	Open a token-based interview link, grant camera/microphone access, and take the interview.
FR-10	System	Generate adaptive, CV/JD-grounded questions and accept spoken or typed answers.
FR-11	System	Run silent proctoring (face count, gaze) and enforce anti-cheating timing during answers.
FR-12	System	Score answers server-side and synthesize a per-candidate interview report.
FR-13	HR	View a candidate's interview report and combined hiring score; regenerate a report on demand.
FR-14	HR	View company-level hiring analytics and per-job pipeline statistics.
FR-15	Candidate	Self-register for candidate practice interviews.

1.6 Non-Functional Requirements

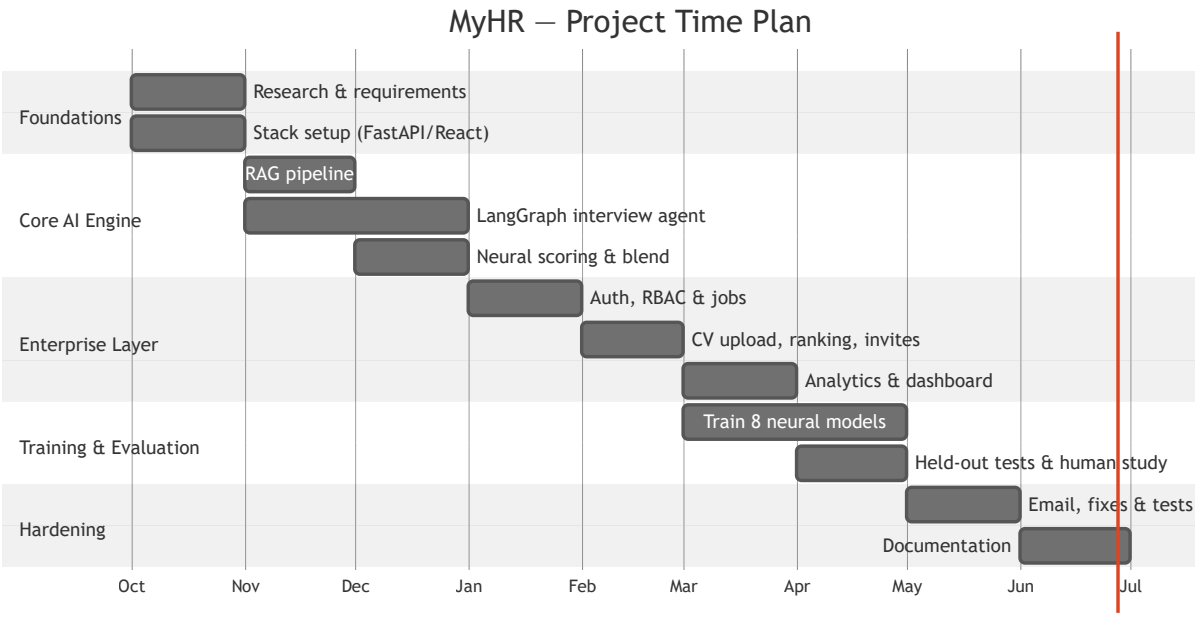
The non-functional requirements define the quality attributes of the system. **Table 1.3** lists them by category.

Table 1.3 — Non-Functional Requirements.

ID	Category	Requirement
NFR-1	Security	All protected endpoints verify a Firebase ID token; scores are computed server-side only.
NFR-2	Security	Public interview links are cryptographically strong, single-purpose, and expire.
NFR-3	Privacy	PII is redacted before indexing; inputs are sanitized against prompt-injection.
NFR-4	Multi-tenancy	Every job-scoped request enforces company ownership (tenant isolation).
NFR-5	Performance	A batch of CVs is parsed and ranked in seconds; analytics are served from pre-computed aggregates.
NFR-6	Reliability	Missing model checkpoints or external services degrade gracefully rather than crash.
NFR-7	Usability	Voice-native interview with a clear status indicator, countdown, and accessible UI.
NFR-8	Maintainability	Clean three-layer separation; core logic covered by an automated test suite.
NFR-9	Portability	Containerized with Docker; runs on CPU or CUDA GPU.
NFR-10	Transparency	Answer scores are an explainable weighted blend, not a single opaque number.

1.7 Project Timeline

Figure 1.1 — Project Time Plan. The project was executed in five overlapping phases: foundations and research, core AI engine, enterprise layer, training and evaluation, and hardening and documentation.



1.8 Documentation Outline

The remainder of this document is organized as follows:

- **Chapter 2 (Background)** reviews the techniques and technologies the system relies on: retrieval-augmented generation, LLM agent orchestration, transformer embeddings and reranking, neural answer scoring, reinforcement learning for adaptive difficulty, emotion recognition and proctoring, and the platform stack.
- **Chapter 3 (System Analysis & Design)** presents the high-level architecture, the three-layer decomposition, the component and deployment diagrams, the database entity-relationship model, and the principal end-to-end workflows as sequence and activity diagrams.
- **Chapter 4 (System Implementation)** describes each component in detail — the RAG pipeline, the interview agent, the neural models, the enterprise layer, proctoring and anti-cheating, the training layer, the database, the API surface, authentication and authorization, and configuration — using diagrams and pseudocode rather than source listings.
- **Chapter 5 (System Testing)** explains the testing strategy, how to install, configure, and run the system, the automated test suite, and the end-to-end golden path with screenshots.
- **Chapter 6 (Results & Discussion)** reports the measured outcomes — model metrics, system performance, strengths, weaknesses, and lessons learned.
- **Chapter 7 (Conclusion and Future Work)** summarizes the achievements, states the known limitations honestly, and proposes future improvements.

The document closes with the tools used, references, a glossary, and appendices.

Chapter Two

Background

Chapter Outline

- 2.1 Retrieval-Augmented Generation
- 2.2 LLM Agent Orchestration
- 2.3 Transformer Embeddings & Reranking
- 2.4 Neural Answer Scoring
- 2.5 Reinforcement Learning for Adaptive Difficulty
- 2.6 Emotion Recognition & Proctoring
- 2.7 Platform Technologies

This chapter introduces the concepts, algorithms, and technologies that MyHR is built upon. It is intended to give the reader the background necessary to follow the system design in Chapter 3 and the implementation in Chapter 4.

2.1 Retrieval-Augmented Generation

A **Large Language Model (LLM)** generates text by predicting the most likely continuation of a prompt. While powerful, an LLM only "knows" what was in its training data and what is placed in its prompt; asked about a specific candidate, it will confidently *hallucinate* details. To ground an LLM in trustworthy, up-to-date facts, **Retrieval-Augmented Generation (RAG)** first *retrieves* the most relevant passages from a source corpus and then *augments* the LLM's prompt with them, so the model reasons over real evidence rather than guesses.

A retrieval system can be **sparse** or **dense**:

- **Sparse retrieval** matches exact terms. **BM25** (Best Matching 25) is the classical sparse ranking function; it scores a document by the frequency of the query's terms within it, discounted by how common those terms are across the corpus. BM25 excels at exact keyword overlap (e.g. a specific framework name) but misses paraphrases.
- **Dense retrieval** matches *meaning*. Each passage is encoded into a high-dimensional vector (an *embedding*) by a transformer model, and relevance is measured by vector similarity (cosine distance). Dense retrieval captures semantic similarity even when no words overlap, but can miss rare exact terms.

Because the two approaches have complementary strengths, modern systems **fuse** them. **Reciprocal Rank Fusion (RRF)** combines several ranked lists into one by summing, for each document, a score of `1 / (k + rank)` across the lists, where `k` is a smoothing constant (commonly 60). RRF is robust because it depends only on a document's *rank* in each list, not on incomparable raw scores. Finally, a **cross-encoder reranker** re-scores the fused shortlist by reading the query and each candidate passage *together*, yielding the most precise ordering at the cost of more computation — which is why it is applied only to the top fused candidates rather than the whole corpus.

MyHR uses exactly this stack: BM25 + dense (Pinecone) retrieval, RRF with `k = 60`, and a cross-encoder reranker, to ground each interview question in the candidate's CV and the JD.

2.2 LLM Agent Orchestration

A single LLM call is stateless. A realistic interview, however, is a *process*: rewrite the context, retrieve evidence, check that the evidence is good enough, generate a question, accept an answer, evaluate it, decide whether to continue, and adapt. Coordinating these steps requires an **agent** — a controller that maintains state and routes execution between specialized steps.

LangGraph models such an agent as a **state graph**: a set of *nodes* (each a function that reads and writes a shared state object) connected by *edges*. Some edges are **conditional** — the next node is chosen at runtime based on the current state (for example, "if the retrieved context is insufficient, re-retrieve; otherwise generate the question"). A **checkpointer** persists the state between turns so a long-running, multi-question interview can resume exactly where it left off.

This graph-based formulation makes the interview logic explicit and inspectable, which is important for a system whose outputs must be defensible. MyHR's interview agent is a LangGraph state machine whose nodes are `rewrite`, `retrieve`, `grade`, `generate`, and `process_answer`.

2.3 Transformer Embeddings & Reranking

The quality of dense retrieval depends entirely on the embedding model. MyHR uses `all-mpnet-base-v2`, a sentence-transformer that maps text to a **768-dimensional** vector. It is a strong general-purpose semantic encoder and is used consistently at both indexing time and query time — a deliberate choice, because using one embedder to *train* a model and a *different* one at inference silently degrades accuracy. The same embedder is therefore used for indexing CV/JD chunks, for the relevance gate in the scoring blend, and for re-encoding training answers so that the trained models see the same representation they will see in production.

A **cross-encoder** differs from the bi-encoder used for retrieval. A bi-encoder embeds the query and the document *independently* and compares the two vectors — fast, because document vectors can be precomputed. A cross-encoder feeds the query and document *together* through the transformer and

outputs a single relevance score — slower, but far more accurate. MyHR applies a cross-encoder only as a final reranking step over the small fused candidate set.

2.4 Neural Answer Scoring

Relying on an LLM as the *sole* judge of an answer has a weakness: the score is an opaque opinion of one model. MyHR therefore complements the LLM with purpose-trained neural networks:

- A **Multi-Layer Perceptron (MLP)** is a feed-forward neural network of fully connected layers with non-linear activations. Given a fixed-length feature vector (here, embeddings of the question and answer), an MLP can be trained to regress a quality score.
- A **multi-head** network shares a common representation but ends in several independent output "heads," each predicting a different target. MyHR's evaluator uses three heads to score an answer's **relevance**, **clarity**, and **depth** separately.

These models are trained with standard supervised learning and evaluated with **rank-aware metrics** — most importantly **Spearman's rank correlation**, which measures whether the model orders answers the same way the ground truth does (rather than matching exact values), and **NDCG** (Normalized Discounted Cumulative Gain) for ranking quality. The final answer score is a transparent weighted blend of the LLM judgment and these neural evaluators.

2.5 Reinforcement Learning for Adaptive Difficulty

A good interviewer adapts: if a candidate answers easily, the questions get harder; if they struggle, the questions get easier. This is naturally framed as a **reinforcement learning (RL)** problem, where an *agent* observes a *state* (the candidate's recent performance), takes an *action* (choose the next question's difficulty), and receives a *reward* (a more informative interview).

Proximal Policy Optimization (PPO) is a widely used, stable policy-gradient RL algorithm that improves the policy in small, clipped steps to avoid destructive updates; **REINFORCE** is the simpler policy-gradient baseline. MyHR's adaptive-difficulty module is trained in a simulated interview environment to map a compact performance state to a difficulty action.

2.6 Emotion Recognition & Proctoring

Two auxiliary signals enrich the interview:

- **Emotion recognition** analyzes the candidate's facial expression and/or vocal tone to estimate affective state during the interview, providing context for the report.

- **Proctoring** runs silently to detect integrity issues — whether a face is present, whether the candidate is looking away, and whether multiple faces appear. MyHR uses **OpenCV** with the lightweight **YuNet** face detector for this, deliberately avoiding heavyweight dependencies. Proctoring observations are aggregated per answer and never shown to the candidate.

2.7 Platform Technologies

MyHR is assembled from established, production-grade components.

Table 2.1 — Platform Technology Stack.

Layer	Technology	Role in MyHR
Frontend	React 18 + Vite 5	Single-page application (SPA) for candidates and HR
Frontend	Radix UI + Tailwind CSS	Accessible component library and styling
Frontend	React Router, TanStack Query	Routing and server-state management
Frontend	Recharts, Framer Motion	Analytics charts and animations
Backend	FastAPI (Python)	REST + WebSocket API server
Backend	Uvicorn	ASGI server runtime
AI Agent	LangGraph + LangChain	Interview agent state machine
LLM	Groq `llama-3.3-70b-versatile`	Question generation and answer evaluation
RAG	LlamaIndex + Pinecone	Dense vector index and retrieval
RAG	`rank_bm25`	Sparse BM25 retrieval
Embeddings	`sentence-transformers` (`all-mpnet-base-v2`)	768-D text embeddings
Neural layer	PyTorch	Eight custom models (scoring, ranking, etc.)
Speech	Deepgram SDK	Speech-to-text and text-to-speech
Proctoring	OpenCV (YuNet)	Silent face/attention detection
Privacy	Microsoft Presidio	PII detection and redaction before indexing
Auth	Firebase Authentication	Identity and ID-token verification
Database	Google Cloud Firestore	Multi-tenant document persistence
Email	Resend API / Gmail SMTP	Invitation and notification delivery
Training	MLflow + TensorBoard	Experiment tracking and metrics
Rate limiting	SlowAPI	Per-route request throttling

The next chapter shows how these technologies are composed into the MyHR architecture.

Chapter Three

System Analysis & Design

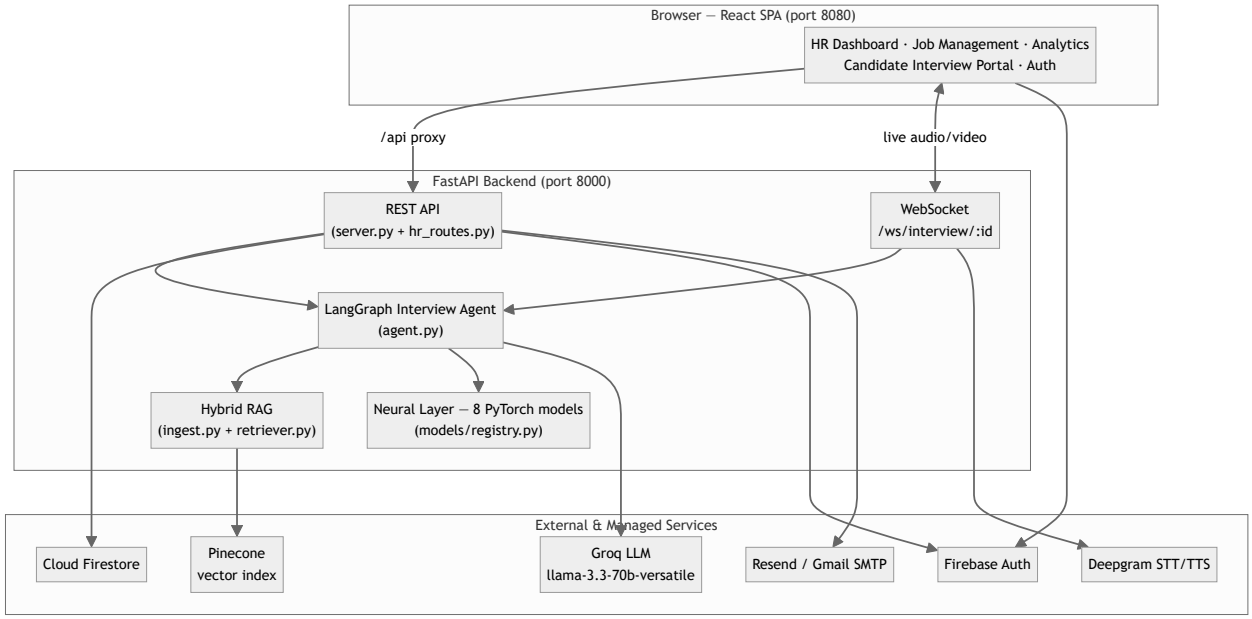
Chapter Outline

This chapter presents the analysis and design of MyHR: how the major components are organized, how they communicate, how users and data flow through the system, and how the persistence, security, and deployment concerns are structured. Design decisions and trade-offs are noted throughout; the detailed implementation of each component follows in Chapter 4.

3.1 System Architecture

MyHR follows a classic **client–server** architecture. A React single-page application runs in the browser and communicates with a FastAPI backend over HTTP (REST) and, during a live interview, over a WebSocket. The backend orchestrates four external/internal services: the LLM (Groq), the vector database (Pinecone), the neural model layer (PyTorch, in-process), and the persistence and identity services (Firestore and Firebase Authentication). Speech conversion is handled by Deepgram, and transactional email by Resend or Gmail SMTP.

Figure 3.1 — System Architecture.

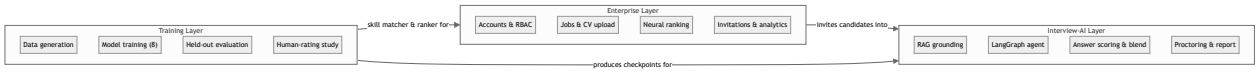


The frontend never talks to Groq, Pinecone, or the neural models directly; all AI work is mediated by the backend, which is also the only place candidate scores are computed.

3.2 Three-Layer Architecture

Conceptually, the system decomposes into three layers with distinct responsibilities, lifecycles, and audiences.

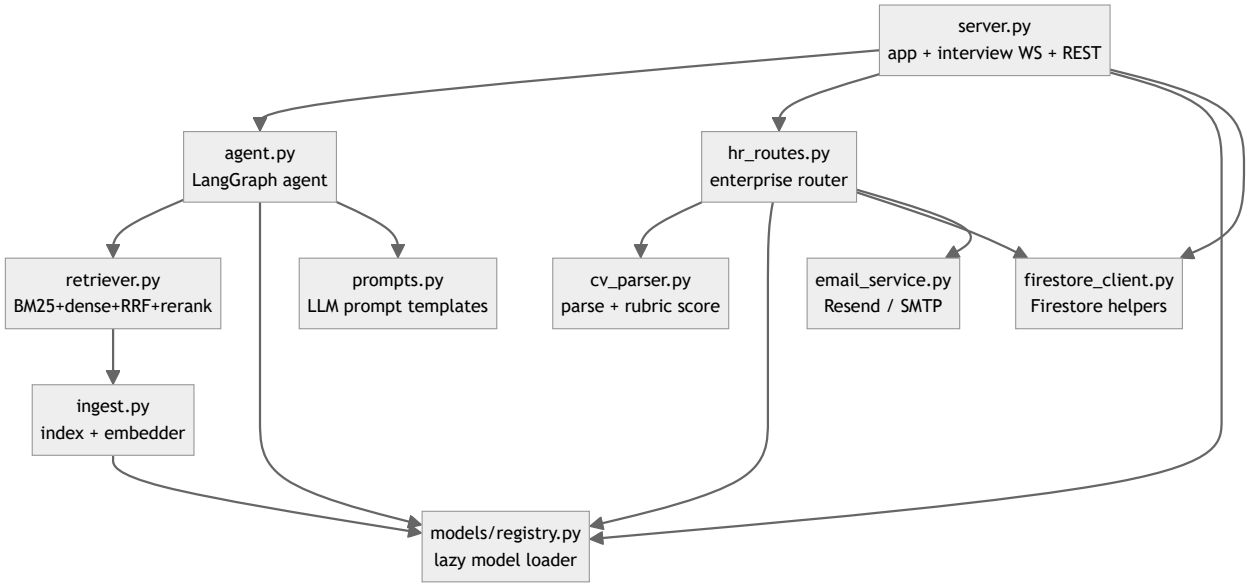
Figure 3.2 — Three-Layer Architecture.



- The **Enterprise Layer** (`hr_routes.py` , frontend HR pages) is multi-tenant, authenticated, and persisted in Firestore. It owns the hiring funnel.
- The **Interview-AI Layer** (`server.py` , `agent.py` , `ingest.py` , `retriever.py` , `models/`) is stateful and real-time. It conducts interviews and scores answers.
- The **Training Layer** (`training/`) is offline. It generates data, trains the eight neural models, evaluates them on held-out test sets, and runs the human-rating validation study. Its outputs are the model checkpoints consumed by the other two layers.

3.3 Component Diagram

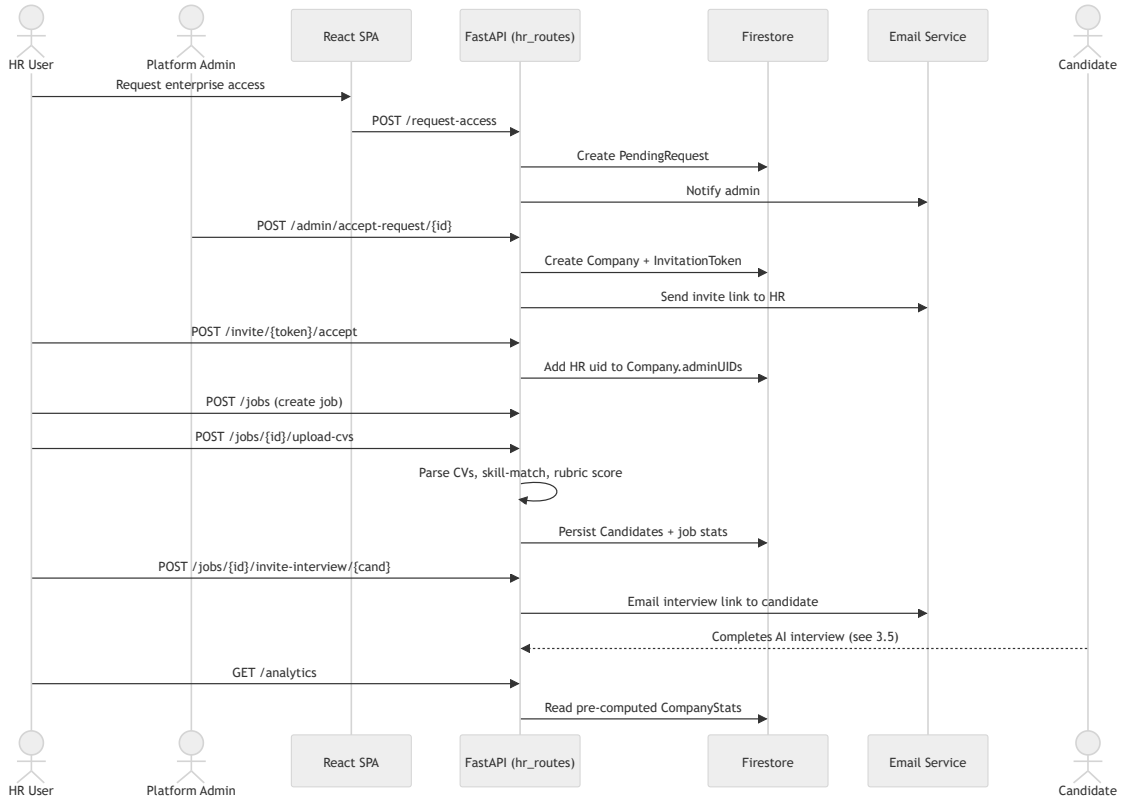
Figure 3.3 — Component Diagram. The following diagram shows the principal backend modules and their dependencies.



3.4 Enterprise Hiring Workflow

The enterprise workflow is the hiring funnel, from a company requesting access through to reviewing completed interviews.

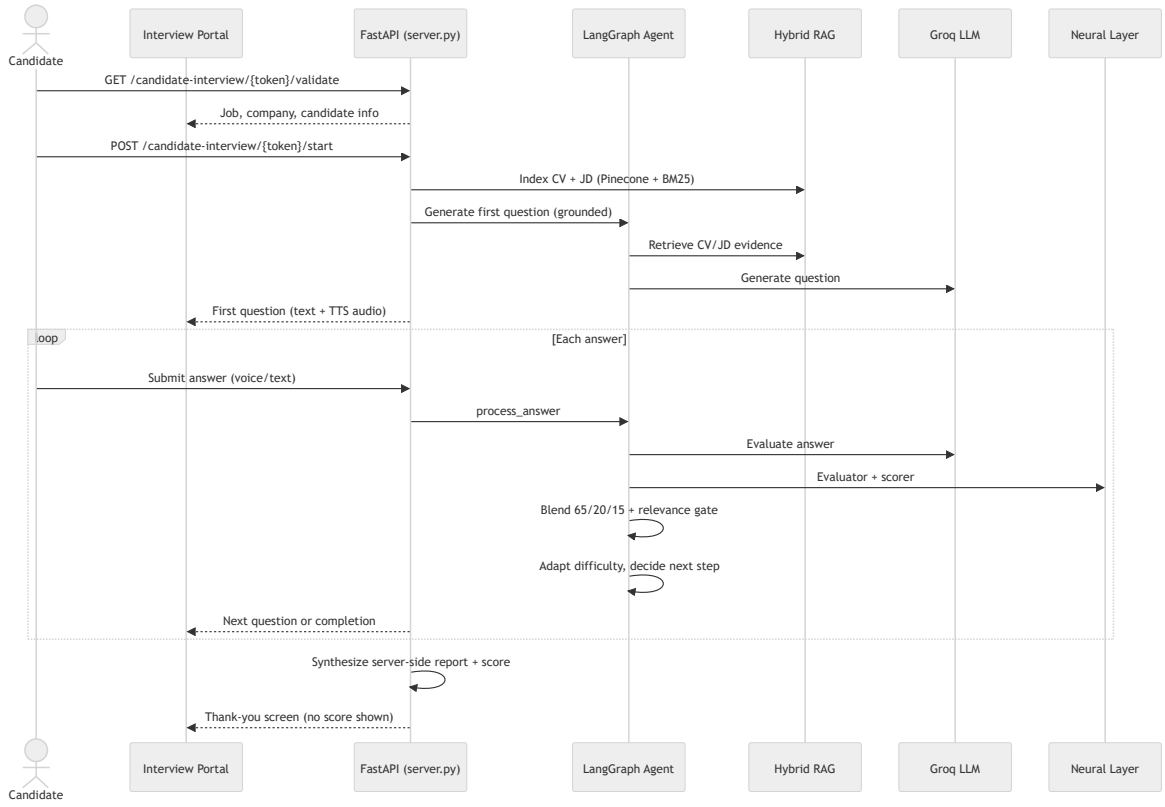
Figure 3.4 — Enterprise Hiring Workflow (Sequence).



3.5 Candidate Interview Workflow

When a candidate opens their interview link, they enter a stateful, real-time session driven by the LangGraph agent.

Figure 3.5 — Candidate Interview Workflow (Sequence).



Two design properties are visible here. First, the question is **grounded** before the LLM is ever called — the agent retrieves CV/JD evidence first. Second, the final score is **synthesized on the server** from the accumulated evaluations; the candidate is shown only a thank-you screen, never their score, and cannot influence the number that reaches the HR dashboard.

3.6 Authentication & Authorization Design

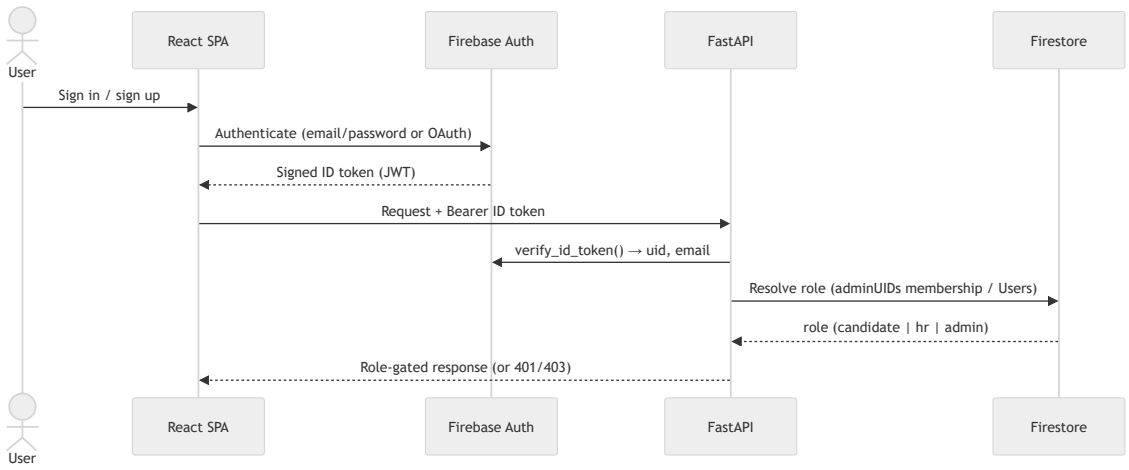
Identity is delegated to **Firebase Authentication**. The browser authenticates the user and receives a signed **ID token** (a JSON Web Token); every protected backend request carries this token in an `Authorization: Bearer …` header, and the backend verifies it with the Firebase Admin SDK before acting. This keeps password handling and token signing entirely within a managed identity provider.

Authorization is **role-based**. Three roles exist and are resolved on the server:

- **Candidate** — self-registerable; may run practice interviews.

- **HR (enterprise)** — *not* self-assignable; granted only by accepting a company invitation, which adds the user's UID to that company's `adminUIDs` array.
- **Platform administrator** — a fixed allowlist (an `ADMIN_EMAIL` plus optional `ADMIN_UIDS`); may approve or reject enterprise access requests.

Figure 3.6 — Authentication & Authorization Flow.



Design rationale. Making the HR role grantable only through an invitation — rather than a self-service toggle — means a malicious user cannot escalate themselves into another company's data. Tenant isolation is then enforced on every job-scoped route by checking that the job's `companyId` matches the caller's company.

3.7 Database Design (Entity-Relationship Model)

MyHR persists all enterprise state in **Cloud Firestore**, a NoSQL document database. The logical entities and their relationships are shown in Figure 3.7; the physical collection layout and field-level detail appear in Chapter 4.6.

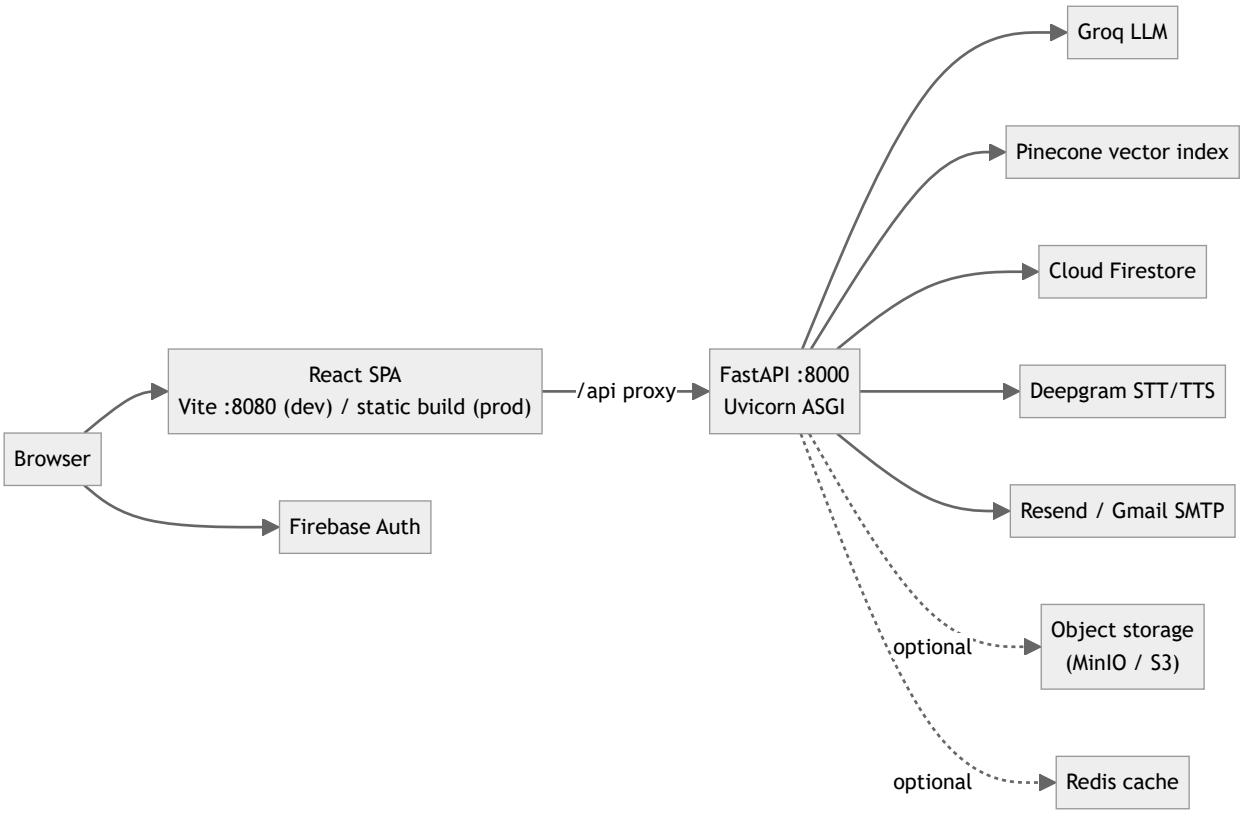
Figure 3.7 — Database Entity-Relationship Diagram.

Design rationale. Candidates are stored as a **subcollection** under each job (`Jobs/{jobId}/Candidates`) rather than in a top-level collection, so that listing a job's candidates is a single scoped query and tenant isolation falls out of the document path. The denormalized `stats` object on each job and the separate `CompanyStats` document trade a small amount of write-time bookkeeping for fast, scan-free dashboard reads.

3.8 Deployment Design

In development the system runs as two processes — the Vite dev server (port 8080) proxying `/api` to the FastAPI backend (port 8000). The project is also containerized (a `Dockerfile` and `docker-compose.yml` are provided) so the backend and its dependencies can be brought up reproducibly.

Figure 3.8 — Deployment Diagram.

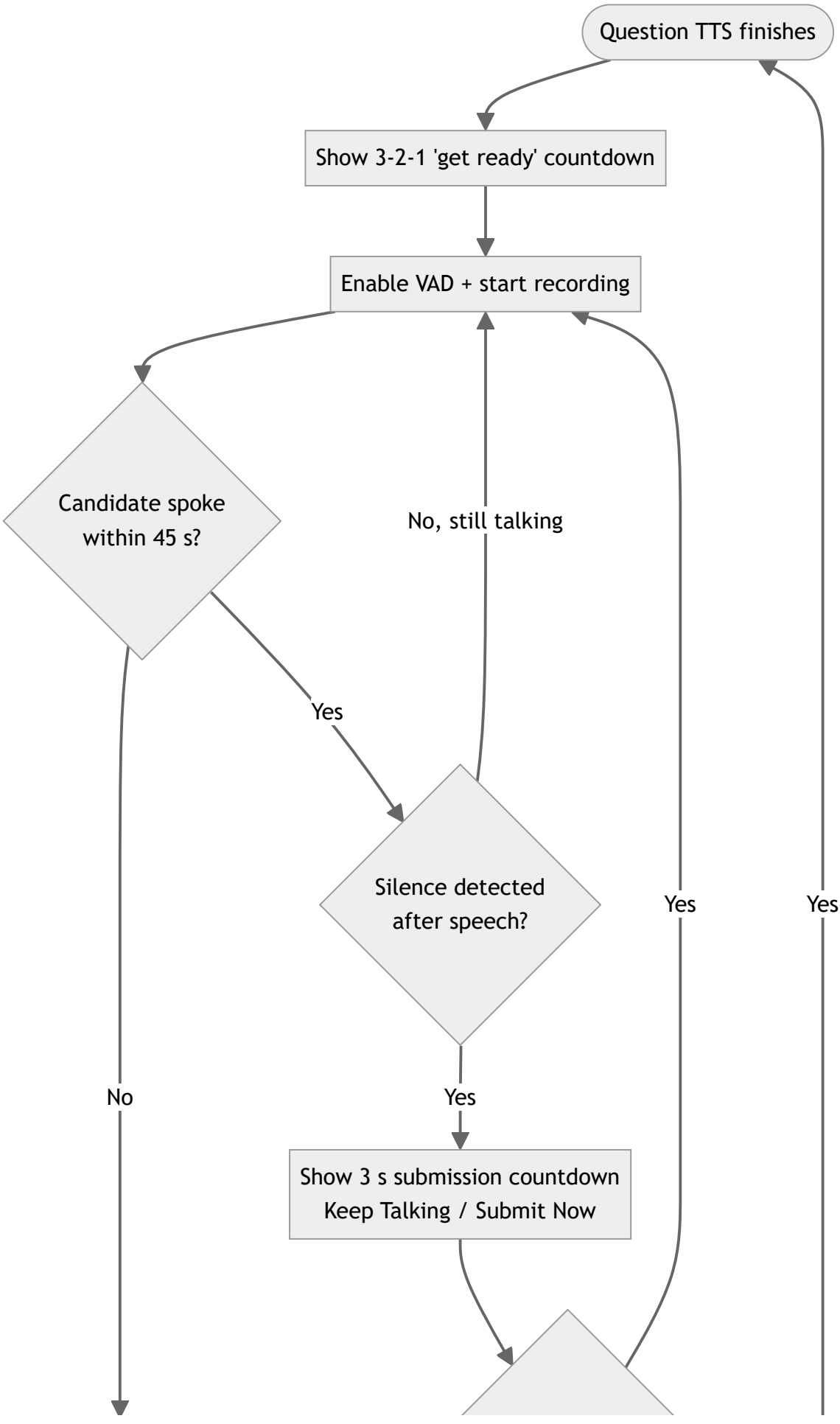


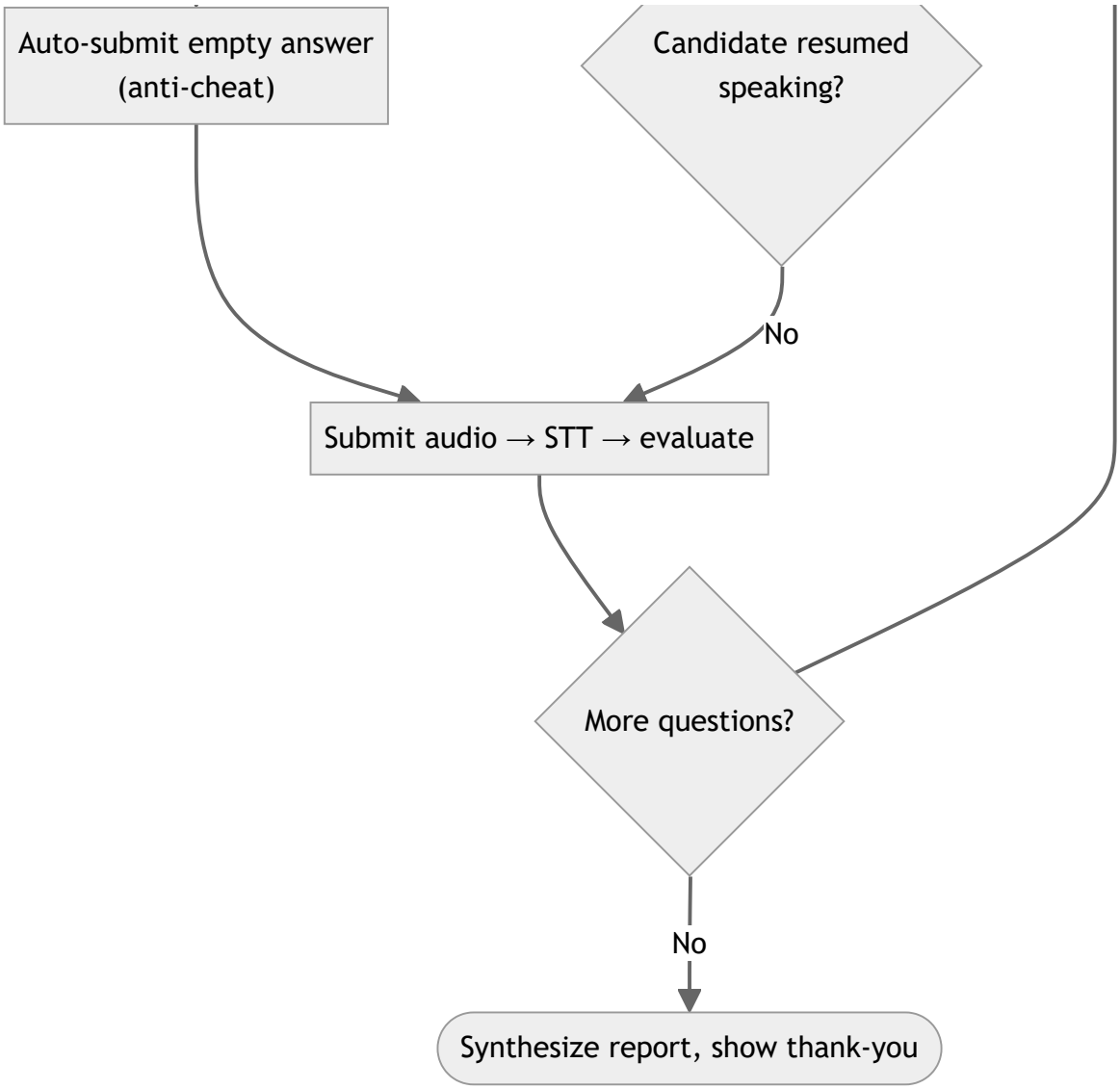
The frontend never contacts Groq, Pinecone, Deepgram, or the neural models directly; all AI work is mediated by the backend, which is the single place candidate scores are computed.

3.9 Interview Activity Diagram

Figure 3.9 shows the control flow of a single interview turn from the candidate's perspective, including the proctoring and anti-cheating timing introduced in the portal.

Figure 3.9 — Interview Turn (Activity Diagram).


Question TTS finishes
Show 3-2-1 'get ready' countdown
Enable VAD + start recording
Candidate spoke within 45 s?
No, still talking
Yes
Silence detected after speech?
Yes
Yes
Yes
No
Show 3 s submission countdown
Keep Talking / Submit Now



The next chapter details how each of these components is implemented.

Chapter Four

System Implementation

Chapter Outline

- 4.1 Hybrid RAG Pipeline
- 4.2 LangGraph Interview Agent
- 4.3 AI/ML Model Layer
- 4.4 Enterprise Layer
- 4.5 Training Layer
- 4.6 Database Design
- 4.7 API Reference
- 4.8 Authentication & Authorization
- 4.9 Configuration
- 4.10 Proctoring, Speech & Anti-Cheating

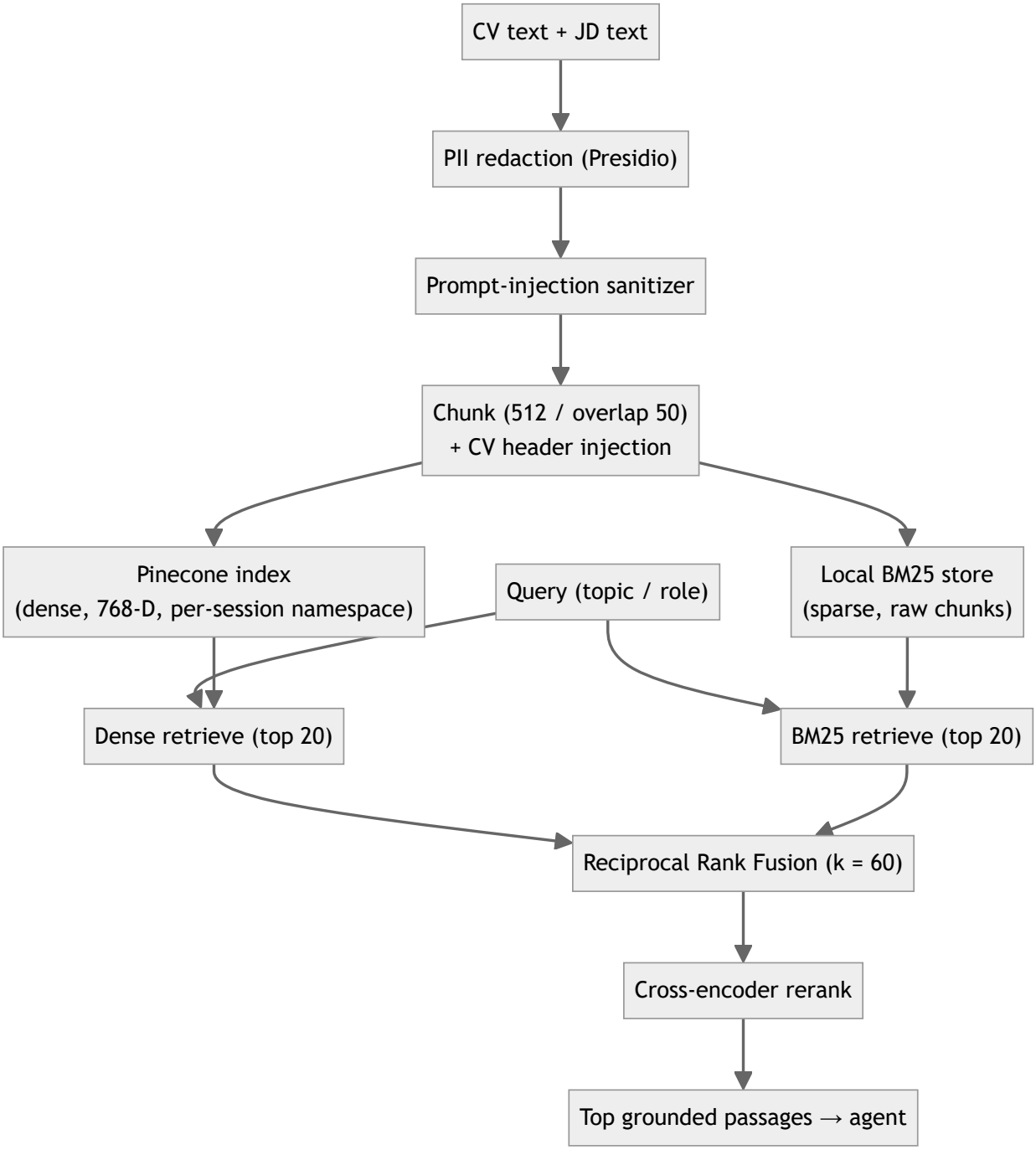
This chapter describes the detailed implementation of each component. In keeping with the documentation standard, it presents **flowcharts and pseudocode** rather than source listings, followed by a short results-and-discussion note for each component.

4.1 Hybrid RAG Pipeline

Modules: `ingest.py` (indexing) and `retriever.py` (retrieval).

When an interview starts, the candidate's CV and the JD are turned into a private, searchable index scoped to the interview session. Before indexing, every document passes through two safety filters: **PII redaction** (Microsoft Presidio replaces names, phones, emails, and similar entities with typed tokens, so raw personal data is never persisted in the vector store) and a **prompt-injection sanitizer** (regular-expression patterns that strip attempts to override the system's instructions). The cleaned text is split into overlapping chunks (`chunk_size = 512` , `chunk_overlap = 50`), each CV chunk is prefixed with a small header carrying the candidate name and role, and the chunks are written to two stores in parallel: **Pinecone** (dense vectors, `all-mpnet-base-v2` , 768-D, namespaced by session) and a **local BM25 store** (raw chunk text for sparse retrieval).

Figure 4.1 — Hybrid RAG Pipeline.



At retrieval time, the query is run against both stores; the two ranked lists are merged with **Reciprocal Rank Fusion** (`score += 1 / (k + rank + 1)` , `k = 60`); and the fused shortlist is re-scored by the cross-encoder reranker for final ordering.

Pseudocode — hybrid retrieval.

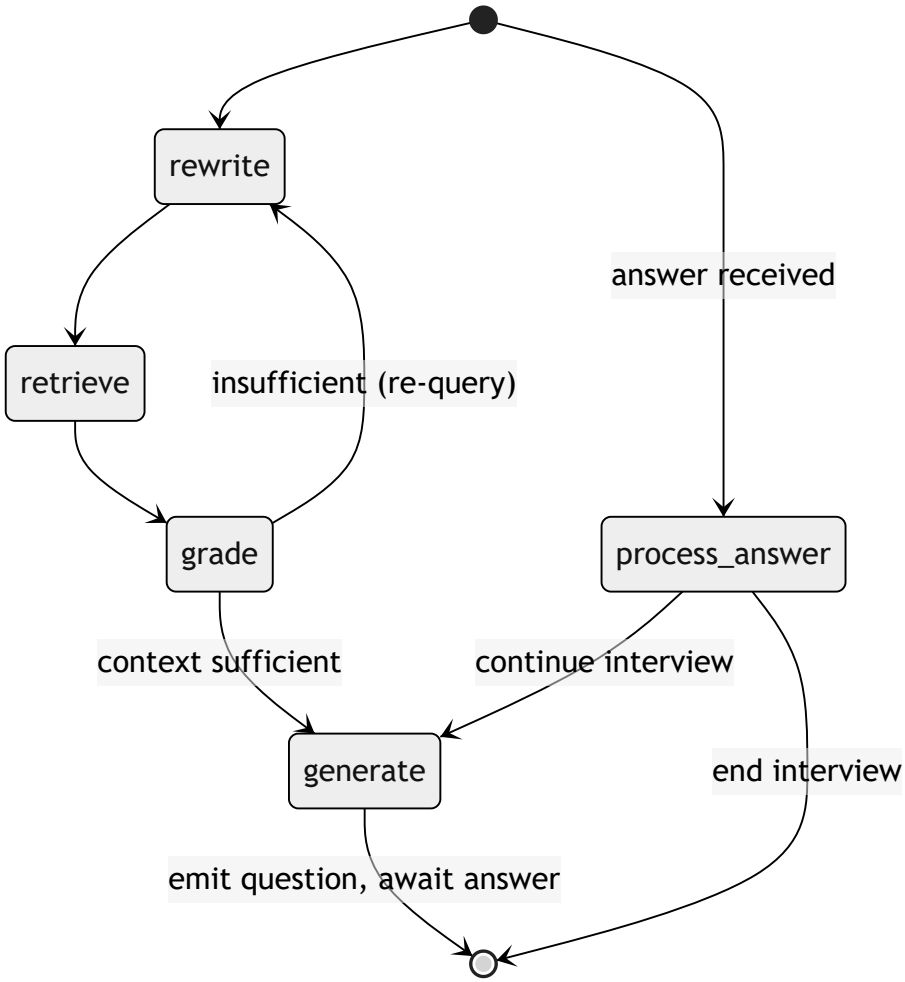
```
function retrieve(query, session_id):
    dense  = pinecone_index(session_id).search(query, top_k = 20)
    sparse = bm25(session_id).search(query, top_k = 20)
    fused  = {}
    for rank, doc in enumerate(dense):  fused[doc] += 1 / (60 + rank + 1)
    for rank, doc in enumerate(sparse): fused[doc] += 1 / (60 + rank + 1)
    shortlist = sort(fused, descending)
    return cross_encoder_rerank(query, shortlist)
```

Results & discussion. The hybrid design retrieves both exact-term matches (BM25 — useful for specific technologies named in a JD) and semantic matches (dense — useful for paraphrased experience). The embedder is initialized lazily at server startup so the ~400 MB model does not block import; the same embedder is used everywhere to avoid train/inference mismatch.

4.2 LangGraph Interview Agent

Module: `agent.py` . The interview is a LangGraph **state graph** compiled with a checkpointer so each session's state survives between turns.

Figure 4.2 — LangGraph Agent State Graph.

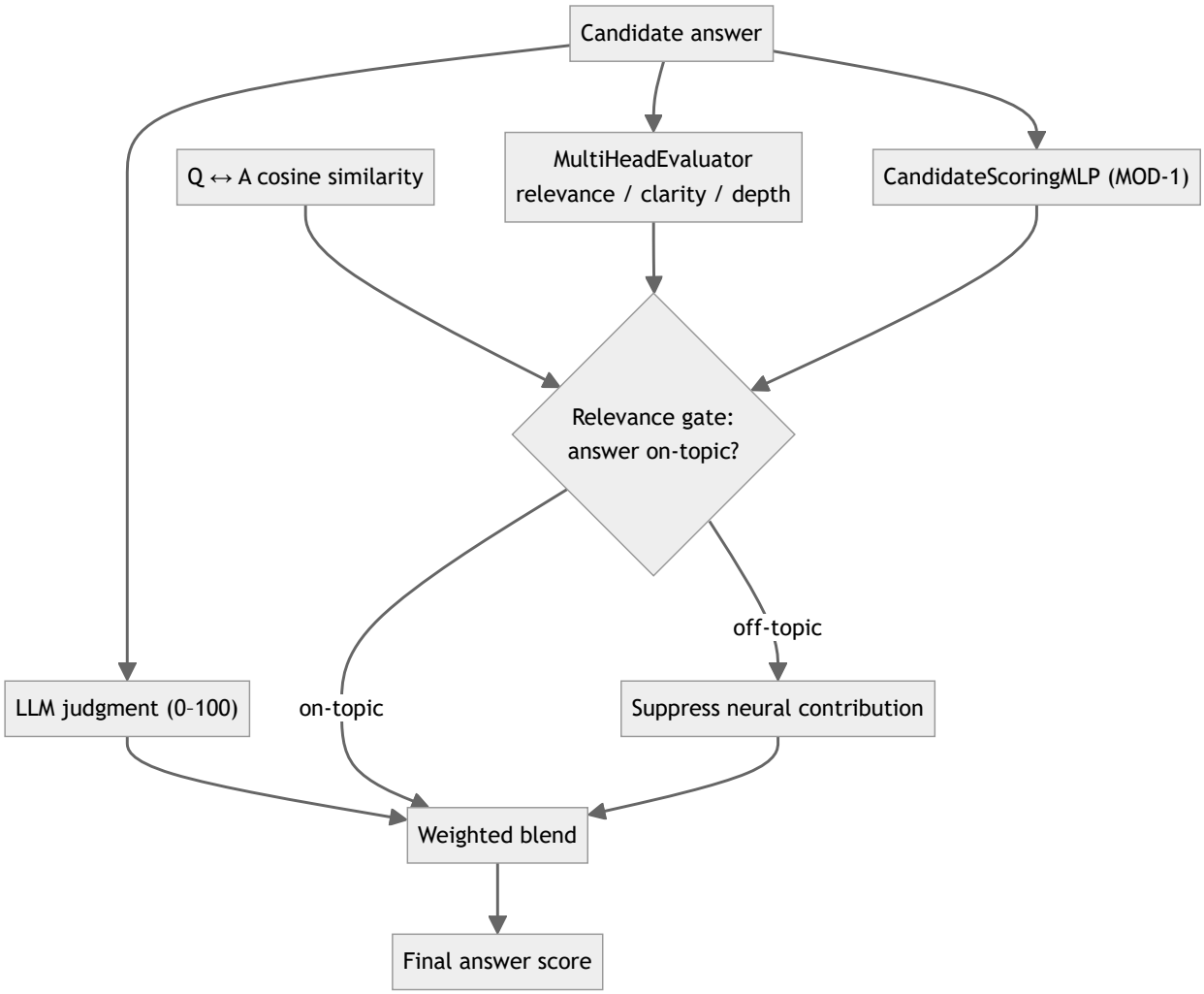


- **rewrite** — reformulates the current topic into an effective retrieval query.
- **retrieve** — runs the hybrid RAG pipeline (4.1) to gather grounded evidence.
- **grade** — checks whether the retrieved context is good enough; a conditional edge (`check_grade`) loops back to re-query if not.
- **generate** — prompts the Groq LLM (`llama-3.3-70b-versatile`) to produce the next question from the grounded context.
- **process_answer** — evaluates the candidate's answer, updates running performance, adapts difficulty, and a conditional edge (`decide_next_step`) decides whether to ask another question or end.

Answer scoring and the blend

When an answer arrives, three independent signals are computed and combined.

Figure 4.3 — Answer Scoring & Blend Flow.



The blend weights are explicit in the code: when all three signals are available, the final score is **65% LLM + 20% neural evaluator + 15% MOD-1** (`_BLEND_3WAY`); when MOD-1 is unavailable, the system falls back to **65% LLM + 35% evaluator** (`_BLEND_2WAY`). A **relevance gate** based on the

cosine similarity between the question and the answer suppresses the neural contribution when the answer is off-topic, preventing a fluent but irrelevant answer from being rewarded by the neural models.

Pseudocode — answer evaluation.

```
function evaluate_answer(question, answer):
    llm   = llm_judge(question, answer)            # 0–100
    eval3 = evaluator(question, answer)            # relevance, clarity, depth
    mod1  = scoring_mlp(question, answer)          # 0–100
    if cosine(question, answer) < threshold:       # relevance gate
        neural = downweighted(eval3, mod1)
    else:
        neural = combine(eval3, mod1)
    if mod1 available:  return 0.65*llm + 0.20*eval_score + 0.15*mod1
    else:               return 0.65*llm + 0.35*eval_score
```

Results & discussion. The blend keeps the LLM as the primary judge while letting purpose-trained models temper its opinion, and the relevance gate guards against the most common failure mode (off-topic fluency). Proctoring signals are aggregated per answer and an answer is flagged "suspicious" if multiple faces appear, the face is absent for more than 20% of frames, or the candidate looks away for more than 30% of frames.

4.3 AI/ML Model Layer

Modules: `models/` with the lazy loader in `models/registry.py`. The registry checks each checkpoint at startup (a health check), but loads each model into memory only on first use, and guards against dimension and embedder mismatches.

Table 4.1 — Neural Model Layer.

#	Model	File	Input	Output	Role
MOD-1	CandidateScoringMLP	`scoring_model.py`	Q+A embeddings (1536-D)	Score 0–100	Deep answer scoring in the blend
MOD-4	MultiHeadEvaluator	`multi_head_evaluator.py`	Answer embedding (768-D)	relevance, clarity, depth	Multi-dimensional answer evaluation
—	SkillMatchSiamese	`skill_matcher.py`	CV skills, JD skills	Match similarity	CV↔JD skill matching for ranking
—	NeuralCandidateRanker	`candidate_ranker.py`	7-D candidate features	Ranking score	Order candidates within a job
—	AdaptiveDifficulty	`difficulty_engine.py`	Performance state (3-D/6-D)	Difficulty action	RL-based question difficulty
—	EmotionModel	`emotion_model.py`	Face/tone features	Emotion estimate	Affective context for the report
—	PerformancePredictor	`performance_predictor.py`	Interview features	Market-positioning estimate	Auxiliary report signal
—	CrossEncoderScorer	`cross_encoder_scorer.py`	(query, passage)	Relevance	RAG reranking

Supporting modules include `proctor.py` (OpenCV YuNet face/attention detection), `feature_extractor.py` (feature assembly), and `explainer.py` (score explanation helpers).

Results & discussion. On held-out test sets (Chapter 4.5), the MultiHeadEvaluator reaches Spearman ≈ 0.95 on each of relevance, clarity, and depth, and the scoring MLP reaches MAE ≈ 0.067 with Spearman ≈ 0.93 — i.e. both models order answers very close to the reference labels. The candidate ranker is currently trained on synthetically generated comparative data and is therefore treated as a *tiebreaker* alongside the rubric and interview scores rather than an absolute measure (see Chapter 7).

4.4 Enterprise Layer

Module: `hr_routes.py` (the `hr_router`), with `cv_parser.py` for CV processing and `firestore_client.py` for persistence.

The enterprise layer implements the hiring funnel. Key implementation details:

- **Batch CV upload** (`POST /jobs/{id}/upload-cvs`) parses each file, extracts contact details and skills, runs the neural **skill matcher** against the JD, computes a **rubric score**, and persists each candidate. Uploads are capped at **5 MiB** per file (`MAX_CV_BYTES`), and duplicate CVs are skipped by content hash and email.
- **Rubric scoring** (`cv_parser.compute_rubric_score`) scores a CV on four axes — technical stack, architecture, experience, and preferred/extra signals — assigns an experience tier (senior/mid/junior), and applies a **knock-out rule** (`framework_cap`): if a CV has zero skill overlap against a JD that lists at least five extractable skills, its score is capped at 40, preventing irrelevant CVs from scoring highly.
- **Atomic job statistics** are updated within a Firestore transaction so concurrent uploads do not lose updates.
- **Pre-computed analytics.** `GET /analytics` reads a pre-computed `CompanyStats` document rather than scanning every candidate on each request; the document is refreshed in the background after CV uploads and interview completions, with an in-memory time-to-live cache as a fast path. This avoids an O(jobs × candidates) scan on every dashboard load.
- **Ownership checks.** Every job-scoped route verifies that the job belongs to the caller's company before returning data, enforcing tenant isolation.

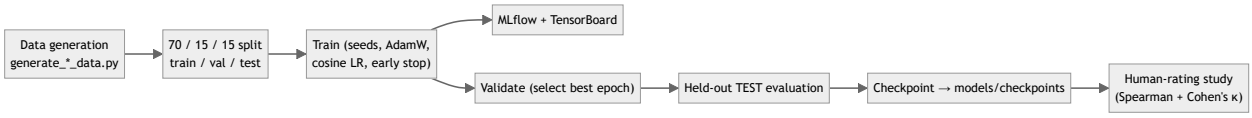
Pseudocode — batch CV upload.

```
function upload_cvs(job_id, files, company_id):
    job = authorize_job(job_id, company_id)
    for file in files:
        if size(file) > 5 MiB: skip
        text   = parse(file)
        if duplicate(hash(text)) or duplicate(email(text)): skip
        skills = extract_skills(text)
        match  = skill_matcher(skills, job.skills)
        score  = rubric_score(text, job.description, job.skills)
        persist_candidate(job_id, {skills, match, score, ...})
    update_job_stats(job_id)         # atomic transaction
    refresh_analytics(company_id)    # background
```

4.5 Training Layer

Module: `training/`. The training layer is offline and produces the checkpoints consumed by the other layers.

Figure 4.4 — Training Pipeline.



Each trainer (`train_evaluator.py` , `train_scorer.py` , `train_ranker.py` , `train_skill_matcher.py` , `train_difficulty.py` / `train_difficulty_ppo.py` , `train_emotion.py` , `train_predictor.py` , `train_cross_encoder.py`) shares the same hygiene: fixed random seeds, an AdamW optimizer with cosine-annealing learning-rate schedule, early stopping with best-state restoration, dual MLflow + TensorBoard logging, and rank-aware metrics (Spearman, NDCG, F1). Models are trained on a **70/15/15** split and final numbers are reported on the **held-out test fold**, not the validation fold used for early stopping.

To prevent a train/inference embedding mismatch, the evaluator's training answers are re-encoded with the same `all-mpnet-base-v2` embedder used in production, and the checkpoint is stamped with the embedder identity so the registry can reject a mismatched model.

Representative held-out TEST metrics.

Model	Metric	Value
MultiHeadEvaluator	Spearman (relevance)	≈ 0.95
MultiHeadEvaluator	Spearman (clarity)	≈ 0.95
MultiHeadEvaluator	Spearman (depth)	≈ 0.95
CandidateScoringMLP	MAE	≈ 0.067
CandidateScoringMLP	Spearman	≈ 0.93

A **human-rating study** (`human_rating_study.py`) compares the system's scores against human ratings on a stratified sample of answers, reporting Spearman's correlation (≈ 0.95) and Cohen's κ (≈ 0.50). A **fairness audit** and a **RAG evaluation** harness are also provided.

Results & discussion. The held-out methodology and consistent embedder give the reported metrics credibility. As noted in Chapter 7, the human study currently has a single rater and should be extended to several raters for inter-rater reliability.

4.6 Database Design

MyHR persists all enterprise state in **Cloud Firestore**, a document database. Collections and their relationships are shown below.

Figure 4.5 — Database Relationships.

Table 4.2 — Firestore Collections.

Collection	Purpose	Key fields
`Companies`	Tenant record	`name`, `adminUIDs`, `domain`, `settings`
`Jobs`	Job postings	`companyId`, `title`, `description`, `extractedSkills`, `stats`
`Jobs/{id}/Candidates`	Candidates per job (subcollection)	`name`, `email`, `matchScore`, `interviewScore`, `totalScore`, `interviewStatus`
`Users`	Portal role registry	`uid`, `role` (`candidate`)
`PendingRequests`	Enterprise access requests	`companyName`, `contactEmail`, `status`
`InvitationTokens`	Access & interview tokens	`type`, `companyId`, `jobId`, `candidateId`, `expiresAt`, `usedAt`
`CompanyStats`	Pre-computed analytics	`stats`, `monthly_trends`, `updatedAt`

The total candidate score combines the CV match and the interview at a fixed weighting — **40% CV match + 60% interview** (`CV_SCORE_WEIGHT = 0.4`, `INTERVIEW_SCORE_WEIGHT = 0.6`).

4.7 API Reference

The backend exposes two routers: interview/system endpoints in `server.py`, and the enterprise router in `hr_routes.py`.

Table 4.3 — API Reference: Interview & System Endpoints (`server.py`).

Method	Path	Auth	Purpose
POST	`/start_interview`	Token/session	Start a (practice) interview session
POST	`/candidate-interview/{token}/start`	Public token	Start a token-based candidate interview
WS	`/ws/interview/{session_id}`	Session	Live audio/video interview channel
GET	`/end_interview/{session_id}`	Session	End an interview and finalize
POST	`/analyze_frame`	Session	Proctoring: analyze a single video frame
POST	`/submit_answer`	Session	Submit and evaluate an answer
POST	`/candidates/rank`	Internal	Rank candidates with the neural ranker
GET	`/health`	Public	Liveness/health check
GET	`/metrics`	Public	In-process metrics

Table 4.4 — API Reference: Enterprise Endpoints (`hr_routes.py`).

Method	Path	Auth	Purpose
POST	`/request-access`	Public	Submit enterprise access request
GET	`/admin/pending-requests`	Admin	List pending requests
POST	`/admin/accept-request/{id}`	Admin	Approve request, create company + invite
POST	`/admin/reject-request/{id}`	Admin	Reject a request
GET	`/invite/{token}/validate`	Public token	Validate a company-access invite
POST	`/invite/{token}/accept`	Firebase	Link user to company (grants HR role)
POST	`/jobs`	HR	Create a job
GET	`/jobs`	HR	List company jobs
GET	`/jobs/{id}`	HR	Get a job
POST	`/jobs/{id}/upload-cvs`	HR	Batch upload + score CVs
GET	`/jobs/{id}/candidates`	HR	List candidates (paged/filtered)
GET	`/jobs/{id}/candidates/{cid}`	HR	Get a candidate (with report)
DELETE	`/jobs/{id}/candidates/{cid}`	HR	Remove a candidate
POST	`/user/role`	Firebase	Self-register as candidate
GET	`/user/role/{uid}`	Public	Resolve a user's role
POST	`/jobs/{id}/invite-interview/{cid}`	HR	Email an interview invite
GET	`/candidate-interview/{token}/validate`	Public token	Validate interview token
POST	`/candidate-interview/{token}/complete`	Public token	Finalize interview (server-scored)
GET	`/analytics`	HR	Company hiring analytics
POST	`/jobs/{id}/rank-candidates`	HR	Neural ranking of candidates

4.8 Authentication & Authorization

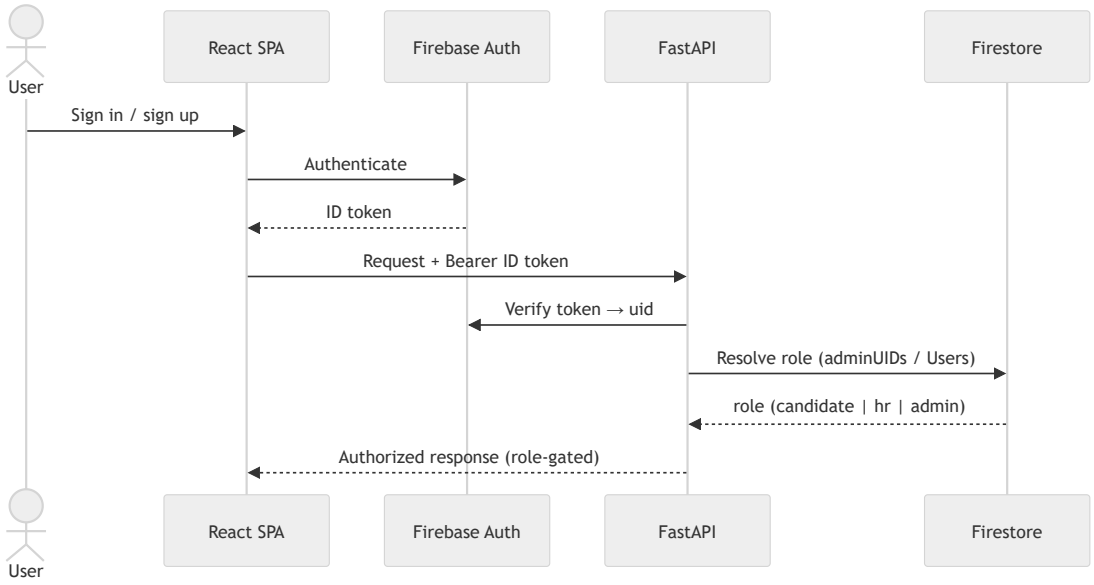
Modules: `firestore_client.py`, `hr_routes.py`, and `src/contexts/AuthContext.jsx` on the frontend. Identity is provided by **Firebase Authentication**; the backend verifies the Firebase **ID token** on protected routes.

Three roles exist:

- **Candidate** — self-registerable (`POST /user/role` accepts only `candidate`). Candidates can run practice interviews.
- **HR (enterprise)** — *not* self-assignable. The role is granted only by accepting a company invitation, which adds the user's UID to the company's `adminUIDs` array; the role resolver (`GET /user/role/{uid}`) reports `hr` precisely when the UID appears in some company's `adminUIDs` . An email already registered as a candidate cannot also become enterprise, and vice-versa — one email maps to one role.
- **Super-admin** — a fixed allowlist email with platform privileges (approving requests).

Public interview links use cryptographically strong tokens (`secrets.token_urlsafe`) with an expiry, validated server-side; the corporate-email gate restricts enterprise sign-up to corporate domains (with a `BYPASS_EMAIL_CHECK` escape hatch for local testing). Interview scores are computed only from the server-side synthesized report, so a candidate cannot submit their own score.

Figure 4.6 — Authentication & RBAC Flow.



4.9 Configuration

Configuration is supplied via environment variables (a local `.env` file in development).

Table 4.5 — Environment Variables.

Variable	Purpose
`GROQ_API_KEY`	Groq LLM access (`llama-3.3-70b-versatile`)
`DEEPGRAM_API_KEY`	Speech-to-text / text-to-speech
`PINECONE_API_KEY`	Pinecone vector index
`OPENAI_API_KEY`	Optional LLM/embedding access
`FIREBASE_SERVICE_ACCOUNT_PATH`	Firebase Admin SDK credentials
`VITE_FIREBASE_*`	Frontend Firebase web config
`RESEND_API_KEY`	Resend email transport
`SMTP_USER` , `SMTP_PASS`	Gmail SMTP transport (App Password)
`MYHR_BASE_URL`	Frontend base URL for email links (e.g. `http://localhost:8080`)
`AWS_*` , `S3_BUCKET_NAME` , `MINIO_ENDPOINT`	Object storage (MinIO/S3) configuration
`REDIS_URL`	Redis connection (caching)
`DATABASE_URL`	Relational database URL (auxiliary)
`BYPASS_EMAIL_CHECK`	Dev escape hatch for the corporate-email gate

The email service tries Gmail SMTP first (when `SMTP_USER` / `SMTP_PASS` are set), then falls back to the Resend API, and logs a warning if neither is configured. The LLM embedder is loaded lazily at startup so model loading does not block application import.

4.10 Proctoring, Speech & Anti-Cheating

Modules: `models/proctor.py` (computer-vision proctoring), the Deepgram integration in `server.py` (speech), and the candidate portal hooks `src/hooks/useVAD.js` and `src/pages/CandidateInterviewPortal.jsx` (anti-cheating).

Computer-vision proctoring

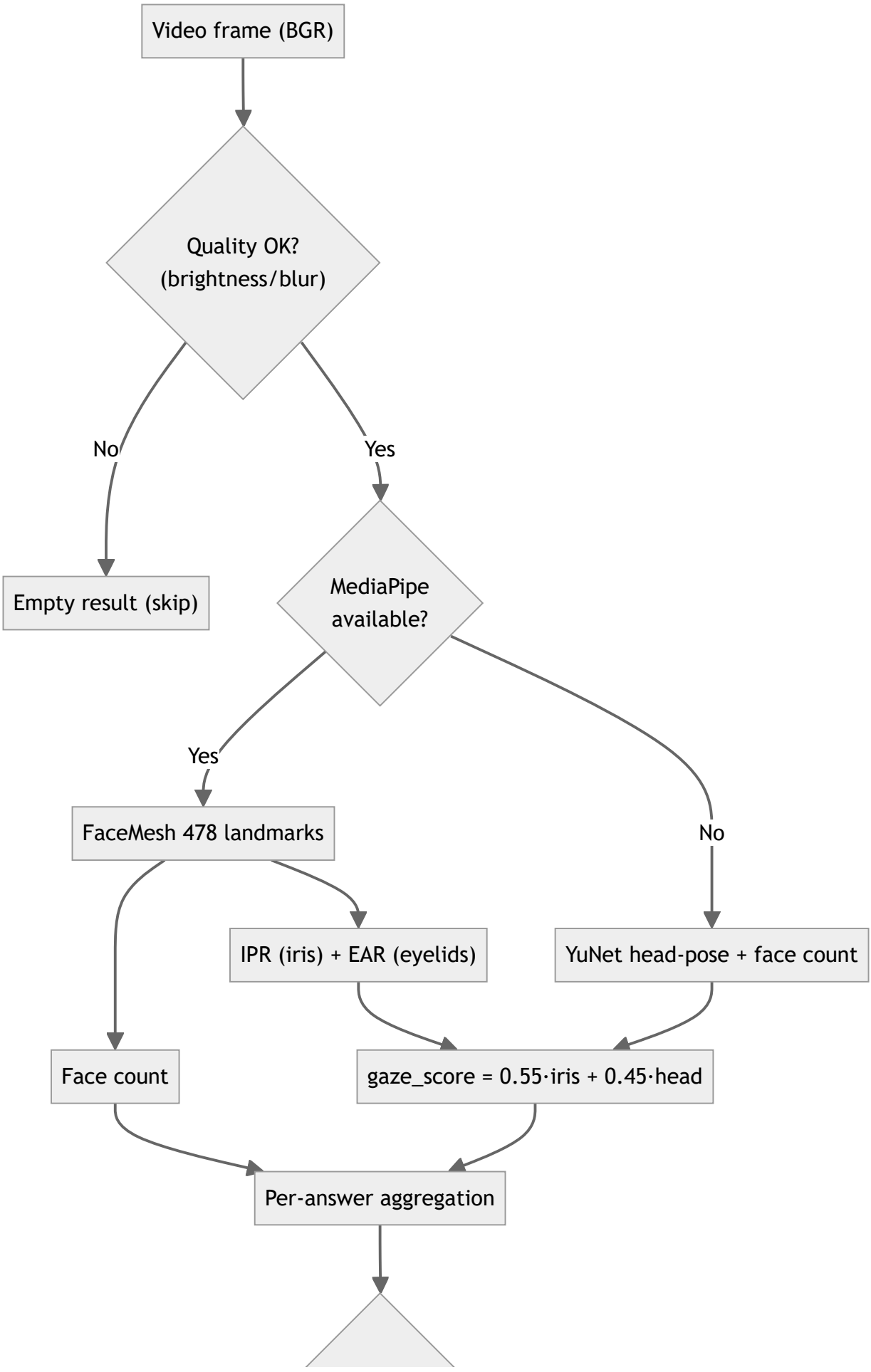
Proctoring runs silently throughout the interview and never surfaces to the candidate. For each captured video frame, the proctor estimates three things: **how many faces are present**, **whether the candidate is looking away**, and **whether the eyes are closed / gaze is extreme**. It uses a three-tier detection chain so it degrades gracefully on machines without the heavier dependency:

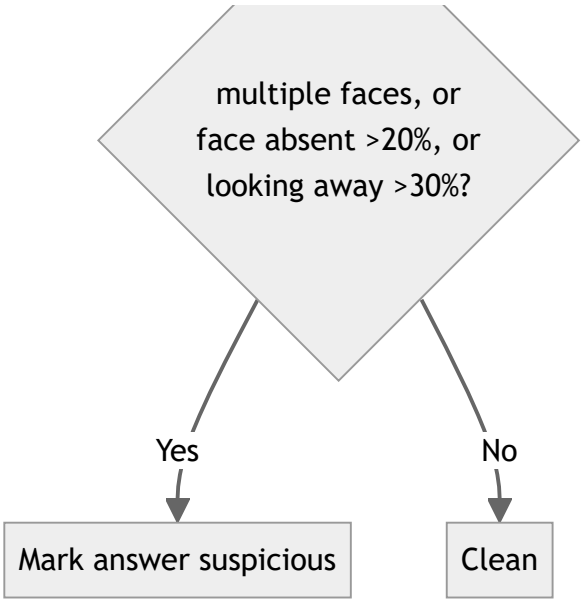
1. **MediaPipe FaceMesh** (`refine_landmarks=True`) provides **478 facial landmarks**, including the iris contours (landmarks 468–477). From these the proctor computes an **Iris Position Ratio (IPR)** — the horizontal position of the iris centre between the eye corners — and an **Eye**

Aspect Ratio (EAR) for closed-eye / extreme-downward-gaze detection. The final gaze score is a weighted combination of iris position (0.55) and head pose (0.45).

2. **YuNet** (`cv2.FaceDetectorYN` , an ONNX detector) provides fast, reliable **face counting** and a head-pose fallback when MediaPipe is unavailable.
3. **Haar cascade** is the final fallback.

Figure 4.7 — Proctoring Detection Pipeline.


Video frame (BGR)
Quality OK? (brightness/blur)
No
Yes
Empty result (skip)
MediaPipe available?
Yes
No
FaceMesh 478 landmarks
IPR (iris) + EAR (eyelids)
YuNet head-pose + face count
Face count
gaze_score = 0.55·iris + 0.45·head
Per-answer aggregation



Observations are **aggregated per answer**. An answer is flagged *suspicious* if multiple faces appear, the face is absent for more than 20% of its frames, or the candidate looks away for more than 30% of its frames. As shown in Chapter 4.2, a suspicious answer incurs a score penalty (−5, or −8 when multiple faces are detected) at blend time.

Speech (STT / TTS)

The interview is voice-native. Each question is synthesized to speech with **Deepgram TTS** and played to the candidate; each spoken answer is captured in the browser, sent to the backend, and transcribed with **Deepgram STT** before being passed to the scoring pipeline. The transcript is what the LLM judge and the neural evaluators actually score, so a candidate may answer by voice or by typing with identical downstream handling.

Anti-cheating in the candidate portal

Two browser-side problems are addressed in the portal. First, **silence is not the same as a finished answer** — a candidate pausing to think must not have an incomplete answer submitted. Second, **a candidate must not be able to skip a question by staying completely silent**. The portal solves both with a small recording state machine built around browser **Voice Activity Detection (VAD)** (Silero VAD via `@ricky0123/vad-web`):

- After a question's audio finishes, a mandatory **3-second "get ready" countdown** is shown before the microphone becomes active. This also makes it impossible to start answering before the question has fully played.
- The VAD's silence tolerance (`redemptionFrames`) is set so that a natural ~250 ms mid-sentence pause does **not** end the answer.
- When the candidate does fall silent after speaking, a **3-second submission countdown** appears with **"Keep Talking"** and **"Submit Now"** controls; speaking again cancels it. Only when the countdown reaches zero (or *Submit Now* is pressed) is the audio submitted.

- If the candidate never speaks at all, a **45-second no-speech timeout** auto-submits an empty answer, so silent non-participation cannot be used to dodge a question.

Figure 4.8 — Anti-Cheating Recording State Machine. *(See also Figure 3.9.)*

```
TTS ends ──▶ [PRE-ANSWER COUNTDOWN 3..2..1] ──▶ [LISTENING]
   [LISTENING] ──speech──▶ [RECORDING] ──silence──▶ [SUBMIT COUNTDOWN 3..2..1]
   [SUBMIT COUNTDOWN] ──speaks again / Keep Talking──▶ [RECORDING]
   [SUBMIT COUNTDOWN] ──reaches 0 / Submit Now──▶ submit
   [LISTENING] ──no speech for 45 s──▶ submit empty (anti-cheat)
```

Results & discussion. Separating "the candidate is thinking" from "the candidate is done" removed the most common usability complaint (premature submission) while the mandatory pre-answer countdown and the no-speech timeout together close the two obvious ways to game a hands-off interview. Because the logic lives in the portal's state, it applies identically to the enterprise candidate portal and the practice interview room.

Chapter Five

System Testing

Chapter Outline

- 5.1 Testing Strategy
- 5.2 Installation
- 5.3 Running the System
- 5.4 Automated Test Suite
- 5.5 End-to-End Walkthrough

This chapter explains the testing strategy, how to install, configure, run, and test MyHR, and walks through the end-to-end golden path with screenshots.

5.1 Testing Strategy

MyHR is tested at several levels, chosen to give the most confidence per unit of effort given a system that mixes deterministic business logic with non-deterministic AI components.

Level	What is tested	How
Unit	Pure logic: CV parsing, skill extraction, rubric scoring, the corporate-email gate, secure token generation, CV-text validation	`pytest` against the functions directly (`test_cv_parser.py` , `test_hr_helpers.py`)
Integration	Cross-module behaviour of the enterprise helpers	`pytest` (`test_integration.py`) with a fake model registry fixture
API	Endpoint contracts (status codes, auth gating, payload shape)	Manual and `pytest` against FastAPI's `/docs` and routes
AI / model	Model quality on held-out test folds (Spearman, NDCG, MAE)	Training-layer evaluators (`training/evaluate_*.py`) reported in Chapter 6
RAG	Retrieval grounding quality	`training/evaluate_rag.py`
Frontend	React component behaviour	**Vitest** + React Testing Library
Manual / UAT	End-to-end golden path (Section 5.5)	Human walkthrough with screenshots

A deliberate design choice is that the **non-deterministic AI is evaluated statistically** (on test sets, by correlation metrics) rather than with brittle exact-output assertions, while the **deterministic logic** (parsing, scoring rules, auth gates) is pinned with fast, exact unit tests. The two heavy concerns — security gating and CV scoring — receive the most unit coverage because they are both deterministic and high-risk. Performance, load, and full security penetration testing are identified as future work (Chapter 7); they are **not applicable in the current implementation** beyond the per-route rate limiting already provided by SlowAPI.

5.2 Installation

MyHR has a Python backend and a Node.js frontend. Both must be installed.

Prerequisites

- Python 3.11+ (the project is currently run on Python 3.14)
- Node.js 18+ and npm
- Accounts/keys for: Groq, Deepgram, Pinecone, Firebase, and an email transport (Resend or a Gmail App Password)

Backend dependencies

```
# from the project root
python -m venv .venv
. .venv/Scripts/activate      # Windows (Git Bash);  source .venv/bin/activate on
Linux/macOS
pip install -r requirements.txt
```

Frontend dependencies

```
npm install
```

Configuration. Create a `.env` file in the project root with the variables listed in **Table 4.5** (Chapter 4.9). At minimum, set `GROQ_API_KEY`, `DEEPGRAM_API_KEY`, `PINECONE_API_KEY`, `FIREBASE_SERVICE_ACCOUNT_PATH`, the `VITE_FIREBASE_*` web config, an email transport (`RESEND_API_KEY` or `SMTP_USER` + `SMTP_PASS`), and `MYHR_BASE_URL` pointing at the frontend (e.g. `http://localhost:8080`).

5.3 Running the System

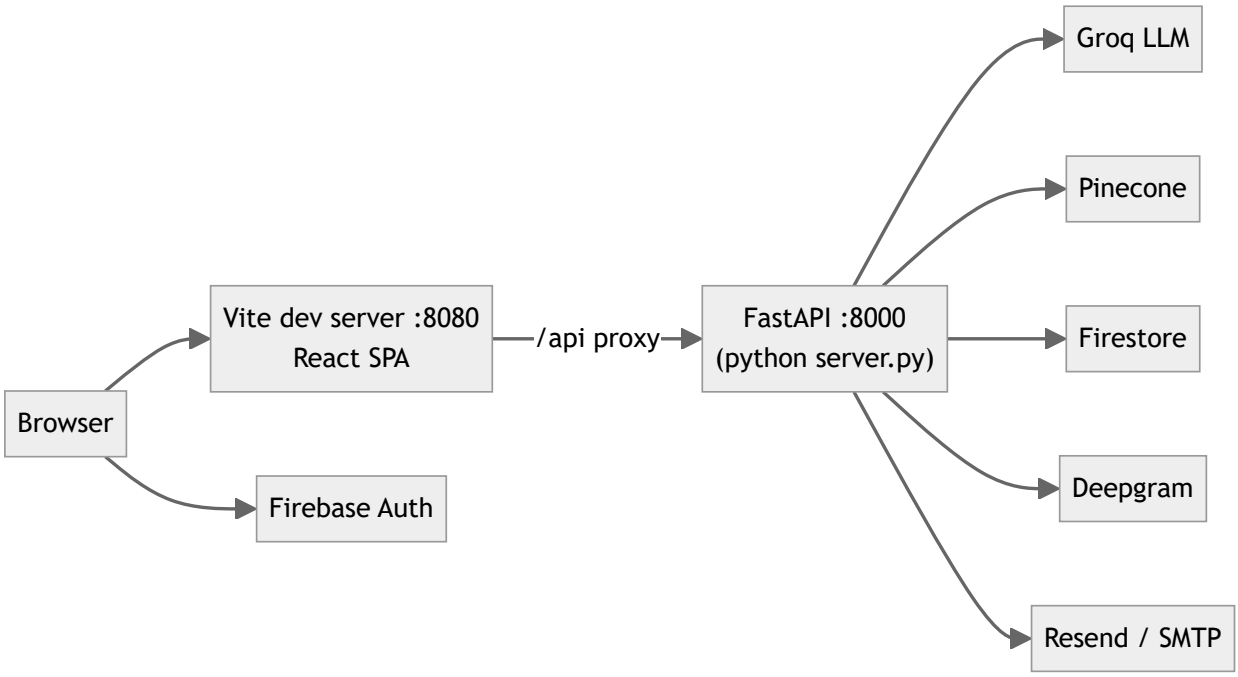
The backend and frontend run as two processes.

```
# Terminal 1 — backend (FastAPI on port 8000)
python server.py

# Terminal 2 — frontend (Vite dev server on port 8080)
npm run dev
```

On a healthy start, the backend logs a model-registry health check (all eight checkpoints present), loads the skill matcher, initializes the LlamaIndex embedder, readies the OpenCV proctor, loads the emotion model, and reports *Application startup complete* with Uvicorn listening on `http://0.0.0.0:8000`. The frontend serves the SPA at `http://localhost:8080`, and Vite proxies all `/api` calls to the backend.

Figure 5.1 — Deployment Architecture.



For a production deployment, the frontend is built with `npm run build` (output in `dist/`) and served as static assets, while the backend runs under a production ASGI server; object storage, caching, and observability would be hardened as described in Chapter 7.

5.4 Automated Test Suite

The backend ships with a `pytest` suite under `tests/`.

```
python -m pytest tests/ -q
```

Table 5.1 — Automated Test Summary.

Test file	Tests	Coverage focus
`tests/test_cv_parser.py`	38	Email/phone/name/skill extraction, skill matching, rubric scoring (incl. the `framework_cap` knock-out rule)
`tests/test_hr_helpers.py`	19	Corporate-email gate, secure token generation, CV-text validation
`tests/test_integration.py`	13	Cross-module integration of enterprise helpers
Total	**70**	All passing

`tests/conftest.py` provides shared fixtures (including a fake model registry) so the suite runs without loading the heavy neural models. The pure-logic helpers (CV parsing, rubric scoring, email validation, token generation) are tested directly, which keeps the suite fast and deterministic.

The frontend uses **Vitest** (`npm run test`) with React Testing Library for component-level tests.

5.5 End-to-End Walkthrough

The following golden path exercises the full system. Screenshot placeholders mark where to insert captured images before submission.

1. Request enterprise access. A company submits the access form; a corporate email is required (unless `BYPASS_EMAIL_CHECK` is set).

![Request access form](file:///d:/Gam3a/Year 4 cs/Graduation Project/MyHR/docs/images/01-request-access.png)

2. Admin approval. The platform admin reviews pending requests and approves one, which creates the company, generates an invitation token, and emails the HR contact a link.

![Admin pending requests](file:///d:/Gam3a/Year 4 cs/Graduation Project/MyHR/docs/images/02-admin-approve.png)

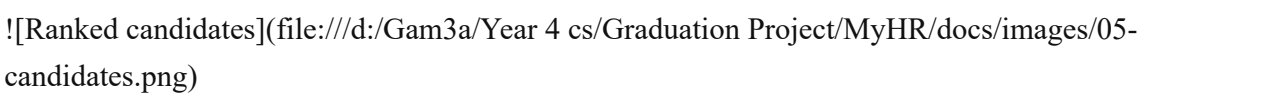
3. Accept invitation. The HR user opens the emailed link, creates their account, is added to the company's `adminUIDs` , and lands on the HR dashboard as an enterprise user.

![Accept invitation](file:///d:/Gam3a/Year 4 cs/Graduation Project/MyHR/docs/images/03-accept-invite.png)

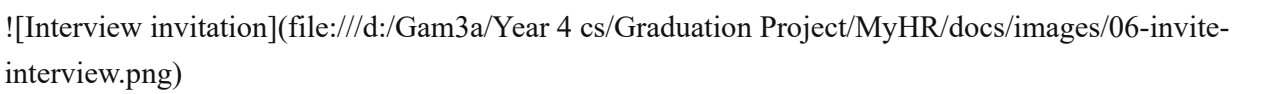
4. Create a job and upload CVs. The HR user posts a job (skills are auto-extracted from the JD) and uploads candidate CVs in bulk; each CV is parsed, skill-matched, and rubric-scored.

![Job management and CV upload](file:///d:/Gam3a/Year 4 cs/Graduation Project/MyHR/docs/images/04-upload-cvs.png)

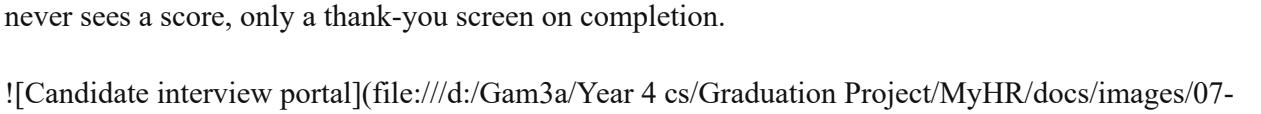
5. Review ranked candidates. Candidates appear ranked by match score; the HR user can open a candidate to see the match breakdown.



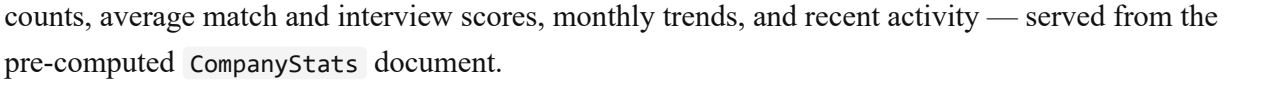
6. Invite a candidate to interview. The HR user invites a candidate; an interview link is emailed to the candidate.



7. Candidate AI interview. The candidate opens the link, grants camera/microphone access, and completes an adaptive, grounded interview (voice or text). Proctoring runs silently; the candidate never sees a score, only a thank-you screen on completion.



8. Hiring analytics. Back on the dashboard, the HR user reviews aggregate analytics — candidate counts, average match and interview scores, monthly trends, and recent activity — served from the pre-computed `CompanyStats` document.



Expected outcome. After completion, the candidate's record shows an `interviewScore`, a synthesized `interviewReport`, and a `totalScore` (40% CV match + 60% interview), and the analytics document refreshes in the background. This confirms the full funnel — from access request to scored interview — operates end-to-end.

Chapter Six

Results & Discussion

Chapter Outline

- 6.1 Model Results (Held-Out Test Sets)
- 6.2 RAG & Grounding Results
- 6.3 System & Backend Performance
- 6.4 Enterprise Funnel Results
- 6.5 Strengths
- 6.6 Weaknesses
- 6.7 Lessons Learned

This chapter reports the measured outcomes of the project and discusses what they mean. Model metrics are reported on **held-out test folds** (the fold used neither for training nor for early-stopping), so they reflect generalization rather than memorization.

6.1 Model Results (Held-Out Test Sets)

The neural layer was trained with a consistent **70 / 15 / 15** train/validation/test split, fixed random seeds, an AdamW optimizer with cosine-annealing learning rate, and early stopping with best-state restoration. Final numbers are reported on the held-out test fold.

Table 6.1 — Model Test-Set Metrics.

Model	Identifier	Primary metric	Value
MultiHeadEvaluator — relevance head	MOD-4	Spearman's ρ	≈ 0.95
MultiHeadEvaluator — clarity head	MOD-4	Spearman's ρ	≈ 0.95
MultiHeadEvaluator — depth head	MOD-4	Spearman's ρ	≈ 0.95
CandidateScoringMLP	MOD-1	Mean Absolute Error	≈ 0.067
CandidateScoringMLP	MOD-1	Spearman's ρ	≈ 0.93

The high Spearman correlations indicate that both models **order** answers very close to the reference labels — which is the property that matters for ranking candidates, more than matching an exact numeric value. The scoring MLP's low MAE (≈ 0.067 on a 0–1 scale) shows the absolute predictions are also well-calibrated.

A **human-rating validation study** (`human_rating_study.py`) compared the system's blended scores against human ratings on a stratified sample of answers, reporting a system-versus-human **Spearman's ρ ≈ 0.95** and **Cohen's κ ≈ 0.50**. The strong rank correlation is encouraging; the moderate κ reflects that exact-category agreement is harder than rank agreement, and is also limited by the study currently using a single rater (see Chapter 7).

Figure 6.1 — Score Distribution & Model Agreement. *(Insert the generated plot from the training run — e.g. the evaluator's predicted-vs-reference scatter and the blend's score-distribution histogram — before submission.)*

6.2 RAG & Grounding Results

The retrieval layer fuses BM25 (sparse) and dense (Pinecone, `all-mpnet-base-v2`) retrieval with Reciprocal Rank Fusion (`k = 60`) and a cross-encoder reranker, evaluated with `training/evaluate_rag.py`. Qualitatively, the most important result is the **grounding guarantee**: because the agent retrieves CV/JD evidence *before* the LLM is prompted, generated questions stay anchored to experience the candidate actually claimed, and the **question-to-answer relevance gate** prevents a fluent but off-topic answer from earning the full neural score. The gate uses the cosine similarity between the question and answer embeddings, scaled so that a similarity of 0.5 or above counts as fully relevant — a ceiling chosen because interview answers rarely exceed ~0.7 similarity even when excellent.

6.3 System & Backend Performance

The system meets its responsiveness objectives (NFR-5):

- **CV screening** turns a batch of uploaded CVs into a ranked shortlist in seconds. Parsing, skill extraction, neural skill matching, and rubric scoring run per file, and job statistics are updated in a single atomic Firestore transaction.
- **Analytics** are served from a pre-computed `CompanyStats` document with an in-memory time-to-live cache, avoiding an O(jobs × candidates) scan on every dashboard load.
- **Model loading** is lazy: the registry verifies checkpoints at startup but loads each model into memory only on first use, and the ~400 MB sentence-transformer embedder is initialized without blocking application import.
- **Graceful degradation** (NFR-6): a missing checkpoint, an unavailable MOD-1, or an absent MediaPipe wheel each fall back rather than crash — the scoring blend drops from 65/20/15 to 65/35, and proctoring drops from iris tracking to head-pose.

Formal load and latency benchmarking of the interview WebSocket under concurrency was not performed and is listed as future work (Chapter 7).

6.4 Enterprise Funnel Results

End-to-end, the platform delivers the complete hiring funnel described in the objectives: a company can move from *requesting access* to a *ranked, interviewed shortlist* without a human reading a single CV or conducting a single first-round interview. Each candidate's record ends with three numbers — a CV `matchScore`, an `interviewScore`, and a combined `totalScore` (**40% CV match + 60% interview**) — plus a synthesized `interviewReport`. Scores are computed server-side only, so the funnel's output is defensible: a candidate cannot influence the number that reaches the HR dashboard.

6.5 Strengths

- **Grounded by construction.** Every question is retrieved-then-generated, so the interviewer cannot interrogate experience the candidate never claimed.
- **Transparent, multi-signal scoring.** The 65/20/15 blend plus relevance gate is explainable and resists the most common gaming strategy (off-topic fluency).
- **Defensible and secure.** Server-side-only scoring, Firebase-verified ID tokens, invitation-only HR role, tenant isolation, PII redaction, and prompt-injection sanitization.
- **Rigorously trained models.** Held-out evaluation, consistent embedder, experiment tracking, and a human-rating study give the metrics credibility.
- **Clean architecture.** Three well-separated layers, 29 documented endpoints, 70 automated tests, and Docker packaging.
- **Robust candidate experience.** Voice-native interview with a usability-tuned, anti-cheat recording state machine.

6.6 Weaknesses

These are scoped for hardening rather than the graduation milestone, and are revisited in Chapter 7:

- **Validation breadth.** Answer-quality labels and the primary judge both originate from an LLM, and the human study uses a single rater — strong agreement is shown, but broader, multi-rater validation would strengthen the claim.
- **Candidate-ranker data.** The neural candidate ranker is trained on synthetically generated comparative data, so it is used as a *tiebreaker* alongside the rubric and interview scores rather than an absolute measure (the code already treats it this way).
- **Storage & observability.** Media/reports use local disk and a MinIO/S3 configuration rather than a hardened cloud bucket, and observability is limited to a health endpoint, an in-process metrics endpoint, and structured logging.

- **No load testing.** Behaviour under concurrent interviews and large uploads has not been benchmarked.

Table 6.2 — System Strengths and Weaknesses (Summary).

Aspect	Strength	Weakness / Limitation
Grounding	Retrieve-then-generate; relevance gate	—
Scoring	Transparent 65/20/15 blend	LLM-origin labels; single-rater human study
Ranking	Neural + rubric + interview	Ranker on synthetic data (tiebreaker only)
Security	Server-side scoring, RBAC, PII redaction	Full pen-testing not performed
Storage	Works locally / MinIO	Not a hardened durable cloud bucket
Observability	Health + metrics + logs	No tracing/alerting/drift monitoring
Performance	Fast screening, cached analytics	No formal load/latency benchmarks

6.7 Lessons Learned

- **Grounding beats raw model size.** The single highest-leverage design decision was forcing retrieval before generation; it did more for answer quality than any prompt tuning.
- **One embedder, everywhere.** Using the same `all-mpnet-base-v2` encoder at training and inference avoided a class of silent accuracy regressions; stamping checkpoints with the embedder identity made mismatches detectable.
- **Silence is ambiguous.** In a voice interview, "the candidate stopped talking" is not the same as "the candidate is finished," and conflating them produced the worst usability bug; separating the two states (with a cancellable submission countdown) fixed it.
- **Defensibility is an architecture property, not a feature.** Keeping all scoring server-side and all role grants invitation-based meant trust did not have to be retrofitted later.

Chapter Seven

Conclusion and Future Work

Chapter Outline

- 7.1 Conclusion
- 7.2 Known Limitations
- 7.3 Future Work

7.1 Conclusion

MyHR set out to automate the two most repetitive stages of hiring — CV screening and first-round interviewing — without sacrificing trustworthiness, and it achieves that goal as a working, end-to-end system.

The project delivers three cooperating subsystems that function together: a **hybrid RAG layer** that grounds every interview question in the candidate's own CV and the job description using BM25, dense Pinecone retrieval, Reciprocal Rank Fusion, and cross-encoder reranking; a **LangGraph LLM agent** that conducts an adaptive, voice-or-text interview and evaluates answers through a transparent blend of an LLM judgment and purpose-trained neural models (65% LLM / 20% evaluator / 15% deep scorer, with a relevance gate); and a **neural layer of eight PyTorch models** trained with professional hygiene — fixed seeds, early stopping, experiment tracking, and reporting on held-out test sets — that reach Spearman correlations around 0.93–0.95 against their reference labels.

Around this AI core, the system provides a complete **multi-tenant enterprise platform**: Firebase-backed authentication, role-based access control that cleanly separates candidates from HR users, batch CV upload with neural skill matching and rubric scoring, email-delivered token-based interview invitations, server-side-only scoring that candidates cannot tamper with, and a pre-computed analytics dashboard. The frontend is a polished React single-page application, the backend a well-structured FastAPI service, and the whole is covered by an automated test suite (70 passing backend tests) and this documentation.

In short, MyHR demonstrates that a grounded, neural-cross-checked AI interviewer can be embedded in a real, secure, multi-tenant hiring product — and that the result is both demonstrable and maintainable.

Principal achievements

- A grounded, adaptive AI interview engine with transparent, defensible scoring.
- Automated CV screening with neural skill matching and a rubric with a knock-out rule.
- A secure, multi-tenant enterprise portal with RBAC and token-based invitations.
- Eight neural models trained and evaluated on held-out test sets with experiment tracking.
- A clean separation of enterprise, interview-AI, and training layers, with automated tests.

7.2 Known Limitations

The system is demo-ready and architecturally healthy. A few areas are deliberately scoped for future hardening rather than the graduation milestone:

- **Validation breadth.** Answer-quality labels and the primary judge both originate from an LLM, so the reported metrics demonstrate strong agreement with that judge; the human-rating validation study, while positive (Spearman ≈ 0.95), currently uses a single rater and would benefit from several raters for inter-rater reliability.
- **Candidate ranker data.** The neural candidate ranker is trained on synthetically generated comparative data, so its scores are best used as a *tiebreaker* alongside the rubric and interview scores rather than an absolute measure. The code already treats it this way.
- **Object storage.** Media and reports are stored against local disk and a MinIO/S3 configuration rather than a hardened, durable cloud bucket.
- **Observability.** The backend exposes a health endpoint, an in-process metrics endpoint, and structured logging, but does not yet include full production monitoring, alerting, or distributed tracing.

None of these affects the demonstrated functionality; each is a natural next step toward a production deployment.

7.3 Future Work

The following improvements would extend MyHR from a strong graduation system toward a production-grade product:

- **Multi-rater validation.** Extend the human-rating study to several independent raters and report inter-rater reliability alongside the system-vs-human correlation.
- **Real comparative labels for ranking.** Retrain the candidate ranker on real hiring outcomes (e.g. which candidates were advanced or hired) to make its scores absolute rather than relative.
- **Durable object storage.** Replace the local-disk path with a hardened cloud object store for CVs, audio, and reports, with lifecycle and access policies.
- **Production observability.** Add metrics dashboards, alerting, request tracing, and model drift monitoring so quality regressions are caught automatically.

- **Model registry and versioning.** Promote the file-based checkpoints to a versioned model registry with rollback, so model updates are auditable and reversible.
- **Scaling and load testing.** Benchmark the interview WebSocket and batch upload paths under load, and introduce horizontal scaling and a continuous-delivery rollback strategy.
- **Richer analytics and fairness reporting.** Surface the existing fairness-audit harness in the dashboard and expand analytics with funnel and time-to-hire metrics.

Pursued in this order, these items would close the gap between the current, defensible demonstration and a system ready for open enterprise use.

Tools

Table T.1 — Tools Used.

Category	Tool	Use in MyHR
Language (backend)	Python 3.14	Backend, AI engine, training
Language (frontend)	JavaScript (ES2022)	React SPA
Web framework	FastAPI + Uvicorn	REST + WebSocket API
Frontend build	Vite 5	Dev server and production bundling
UI	React 18, Radix UI, Tailwind CSS	Component-based interface
Charts/UX	Recharts, Framer Motion	Analytics charts, animations
AI agent	LangGraph, LangChain	Interview state machine
LLM	Groq `llama-3.3-70b-versatile`	Question generation, answer judging
Vector DB	Pinecone (via LlamaIndex)	Dense retrieval
Sparse retrieval	`rank_bm25`	BM25 retrieval
Embeddings	`sentence-transformers` (`all-mpnet-base-v2`)	768-D embeddings
Deep learning	PyTorch	Eight neural models
Speech	Deepgram SDK	STT / TTS
Computer vision	OpenCV (YuNet)	Silent proctoring
Privacy	Microsoft Presidio	PII redaction
Auth	Firebase Authentication	Identity, ID tokens
Database	Google Cloud Firestore	Persistence
Email	Resend API / Gmail SMTP	Invitations, notifications
Experiment tracking	MLflow, TensorBoard	Training metrics
Rate limiting	SlowAPI	Per-route throttling
Testing	pytest, Vitest, React Testing Library	Automated tests
Version control	Git / GitHub	Source management
IDE	Visual Studio Code	Development

Hardware. Development and training used a CUDA-capable GPU workstation (the backend logs `device_name: cuda` when loading sentence-transformer models); the system also runs on CPU.

References

> Books and papers first (ordered by date), then websites (with last-retrieved dates). Replace or extend with the exact sources cited in the final printed document.

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13. Deepgram Documentation, https://developers.deepgram.com/ , last retrieved 27/06/2026.

Glossary & Abbreviations

Term	Definition
API	Application Programming Interface — the contract the backend exposes to the frontend.
ATS	Applicant Tracking System — traditional, keyword-based hiring software.
BM25	Sparse ranking function scoring documents by term overlap with the query.
Bi-encoder	Embeds query and document independently for fast similarity search.
Cross-encoder	Scores a (query, document) pair jointly for higher-precision reranking.
CV	Curriculum Vitae — a candidate's résumé.
Dense retrieval	Semantic search using vector embeddings and cosine similarity.
Embedding	A fixed-length numeric vector representing the meaning of text.
JD	Job Description.
Knock-out rule	A rubric cap that limits the score of a clearly unqualified CV.
LLM	Large Language Model.
MLP	Multi-Layer Perceptron — a feed-forward neural network.
Multi-head	A network with several output heads predicting different targets.
NDCG	Normalized Discounted Cumulative Gain — a ranking-quality metric.
PII	Personally Identifiable Information.
PPO	Proximal Policy Optimization — a reinforcement-learning algorithm.
Proctoring	Silent integrity monitoring (face presence, gaze, multiple faces).
RAG	Retrieval-Augmented Generation.
RBAC	Role-Based Access Control.
Relevance gate	A check that suppresses the neural score when an answer is off-topic.
RL	Reinforcement Learning.
RRF	Reciprocal Rank Fusion — merges several ranked lists into one.
SPA	Single-Page Application.
Spearman's ρ	Rank correlation measuring whether two orderings agree.
STT / TTS	Speech-to-Text / Text-to-Speech.
WebSocket (WS)	A persistent two-way connection used for the live interview.

Appendices

Appendix A — Environment Variable Template

```
# LLM & AI services
GROQ_API_KEY=
DEEPGRAM_API_KEY=
PINECONE_API_KEY=
OPENAI_API_KEY=

# Firebase (Admin SDK + web config)
FIREBASE_SERVICE_ACCOUNT_PATH=
VITE_FIREBASE_API_KEY=
VITE_FIREBASE_AUTH_DOMAIN=
VITE_FIREBASE_PROJECT_ID=
VITE_FIREBASE_STORAGE_BUCKET=
VITE_FIREBASE_MESSAGING_SENDER_ID=
VITE_FIREBASE_APP_ID=

# Email transport (use one)
RESEND_API_KEY=
SMTP_USER=
SMTP_PASS=

# URLs & infrastructure
MYHR_BASE_URL=http://localhost:8080
AWS_ACCESS_KEY_ID=
AWS_SECRET_ACCESS_KEY=
AWS_REGION=
S3_BUCKET_NAME=
MINIO_ENDPOINT=
REDIS_URL=
DATABASE_URL=

# Dev escape hatch (omit in production)
BYPASS_EMAIL_CHECK=
```

Appendix B — Endpoint Index

See **Tables 4.3 and 4.4** for the complete REST/WebSocket surface (9 interview/system endpoints in `server.py`, 20 enterprise endpoints in `hr_routes.py`).

Appendix C — Repository Structure (selected)

```
MyHR/
├── server.py              FastAPI app, interview WebSocket, system endpoints
├── hr_routes.py           Enterprise router (jobs, CVs, invites, analytics)
├── agent.py               LangGraph interview agent + scoring blend
├── ingest.py              RAG indexing + lazy embedder
├── retriever.py           BM25 + dense + RRF + cross-encoder rerank
├── cv_parser.py           CV parsing, skill extraction, rubric scoring
├── email_service.py       Resend / Gmail SMTP transport + templates
├── firestore_client.py    Firestore helpers + role sync
├── prompts.py             LLM prompt templates
├── models/                8 neural models + registry + proctor
│   ├── registry.py
│   ├── multi_head_evaluator.py, scoring_model.py, candidate_ranker.py
│   ├── skill_matcher.py, difficulty_engine.py, emotion_model.py
│   ├── performance_predictor.py, cross_encoder_scorer.py, proctor.py
│   └── checkpoints/       Trained model weights
├── training/              Data generation, trainers, evaluation, human study
├── tests/                 pytest suite (70 tests)
├── src/                   React SPA (pages, components, contexts, hooks, lib)
└── docs/                  This documentation set
```

> **Note on auxiliary modules.** The repository also contains supporting modules not central to the core flow described above — for example `tone.py`, `services.py`, `s3_utils.py`, `feature_extractor.py`, `explainer.py`, and the `recommender/` package — as well as utility and demo scripts (`run_training.py`, `check_phase1.py`, `demo_phase1.py`, `cleanup_pinecone.py`). These are part of the codebase but are outside the primary request-lifecycle documented in Chapter 4.